STA 6930
Bayesian Data Analysis
Lecture Notes

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1 Belief, probability, and exchangeability

1.1 Review of concepts

Sampling space $\mathcal{Y}$: $y \in \mathcal{Y}$
Parameter space Θ: $\theta \in \Theta$

In classic/frequentist statistics: Y is random, $p(y \mid \theta)$ denotes the distribution or likelihood, while θ is fixed.

In Bayesian statistics: θ is random and its probabilistic distribution is called a prior distribution $p(\theta)$.

The Bayes Rule

$$p(\theta \mid y) = \frac{p(y \mid \theta)p(\theta)}{\int_\Theta p(y \mid \theta)p(\theta)d\theta}$$

Example (BDA, p.8) Human males have one X-chromosome and one Y-chromosome, whereas females have two X-chromosomes, each chromosome being inherited from one parent. Hemophilia is a disease that exhibits X-chromosome-linked recessive inheritance, meaning that a male who inherits the gene that causes the disease on the X-chromosome is affected, whereas a female carrying the gene on only one of her two X-chromosomes is not affected. The disease is generally fatal for women who inherit two such genes, and this is rare, since the frequency of occurrence of the gene is low in human populations. Consider a woman who has an affected brother. What is the probability that the woman is a carrier of the gene, given that both of her two sons are unaffected? How about her third son is also not affected?

Let $\{H_1, \ldots, H_k\}$ be a partition of space and E be some event, then

$$P(H_j \mid E) = \frac{P(E \mid H_j)P(H_j)}{\sum_k P(E \mid H_k)P(H_k)}$$

The notation H_j often refers to a hypothesis and E refers to an event or evidence. To compare (test) hypotheses, we define the Bayes factor

$$\begin{aligned} \frac{P(H_i \mid E)}{P(H_j \mid E)} &= \frac{P(E \mid H_i)}{P(E \mid H_j)} \times \frac{P(H_i)}{P(H_j)} \\ \text{Posterior odds} &= \text{Bayes factor} \times \text{Prior odds} \end{aligned}$$

1.2 Review of random variable

Discrete outcome Y (countable):
Probability mass function (pmf): $0 \leq p(y) \leq 1$, $\sum_y p(y) = 1$.
Probability: $P(Y \in A) = \sum_{y \in A} p(y)$.
Example: Poisson distribution, Binomial distribution

Continuous outcome Y:
Probability density function (pdf): $p(y) \geq 0$, $\int_{-\infty}^{\infty} p(y)dy = 1$.
Probability: $P(Y \in A) = \int_A p(y)dy$.
Example: Normal distribution, Gamma distribution

Mean (or expectation):
$\mathrm{E}(Y) = \sum_y yp(y)$ if discrete,
$\mathrm{E}(Y) = \int_y yp(y)dy$ if continuous.

Variance: $\mathrm{Var}(Y) = \mathrm{E}[Y - \mathrm{E}(Y)]^2$.

1.3 Joint distribution

Let Y_1 and Y_2 denote two random variables. The pmf of two discrete random variables is defined as

$$p(y_1, y_2) = P\left\{(Y_1 = y_1) \cap (Y_2 = y_2)\right\}$$

and the marginal pmf is

$$p(y_1) = P(Y_1 = y_1) = \sum_{y_2} p(y_1, y_2)$$

With that, we can define the conditional pmf:

$$p_{Y_2|Y_1}(y_2 \mid Y_1 = y_1) = \frac{p(y_1, y_2)}{p(y_1)}$$

The pdf of two continuous random variables Y_1 and Y_2 is denoted by $p(y_1, y_2)$. The marginal density is given by

$$p(y_1) = \int_{-\infty}^{\infty} p(y_1, y_2) dy_2$$

and the conditional density is

$$p(y_2 \mid y_1) = \frac{p(y_1, y_2)}{p(y_1)}$$

To calculate a probability

$$P(Y_1 \in A, Y_2 \in B) = \int_{y_2 \in B} \int_{y_1 \in A} p(y_1, y_2) dy_1 dy_2$$

Example: $p(y_1, y_2) = 4y_1y_2, 0 < y_1 < 1, 0 < y_2 < 1$.

Question: how should the joint, marginal, and conditional distributions be defined for mixed discrete and continuous random variables?

The Bayes Rule for y and θ (The most fundamental idea of the entire field of Bayesian statistics!)

$$P(\theta \mid y) = \frac{p(y \mid \theta)p(\theta)}{p(y)}$$

where $p(y) = \int_{\theta \in \Theta} p(y \mid \theta)p(\theta)d\theta$. We often write

$$P(\theta \mid y) \propto p(y \mid \theta)p(\theta),$$

due to the left-hand side being a distribution about θ, and therefore $p(y)$ with respect to θ is a constant, referring to as the normalizing constant.

1.4 Independent random variables

Random variables $Y_1, \ldots, Y_n$ are independent if

$$p(y_1, \ldots, y_n \mid \theta) = p(y_1 \mid \theta)p(y_2 \mid \theta) \cdots p(y_n \mid \theta)$$

When the sampling distributions for them are the same, we say that $Y_1, \ldots, Y_n$ are independent and identically distributed (i.i.d.) and

$$p(y_1, \ldots, y_n \mid \theta) = \prod_{i=1}^{n} p(y_i \mid \theta)$$

denoted as $Y_1, \ldots, Y_n \mid \theta \overset{\text{i.i.d.}}{\sim} p(y \mid \theta)$.

Definition 1. *(Exchangeability) Let $p(y_1, \ldots, y_n)$ be the joint density of $Y_1, \ldots, Y_n$. If*

$$p(y_1, \ldots, y_n) = p(y_{\pi_1}, \ldots, y_{\pi_n})$$

for all permutations π of $\{1, 2, \ldots, n\}$, then $Y_1, \ldots, Y_n$ are exchangeable.

Relationship of i.i.d. and exchangeability

Theorem 1. *If $\theta \in p(\theta)$ and $Y_1, \ldots, Y_n$ are i.i.d. given θ, then marginally $Y_1, \ldots, Y_n$ are exchangeable.*

So i.i.d. leads to exchangeability, what about the reverse?

Theorem 2. (De Finetti) *Let $Y_i \in \mathcal{Y}$ for all $i \in \{1, 2, \ldots\}$. Suppose that, for any n, our belief model for $Y_1, \ldots, Y_n$ are exchangeable. Then, our model can be written as*

$$p(y_1, \ldots, y_n) = \int \left\{ \prod_{i=1}^{n} p(y_i \mid \theta) \right\} p(\theta) d\theta$$

for some θ with $p(\theta)$ and some $p(y \mid \theta)$.

De Finetti's theorem establishes the equivalence between the frequentist i.i.d. assumption and the Bayesian exchangeability assumption in a certain way.

2 Single-parameter models

2.1 The Binomial model

Motivating example: Sex ratio (BDA, p.29). The sampling model: $Y_i \mid \theta \sim$ Bernoulli(θ). The model implies the likelihood function

$$p(y_1, \ldots, y_n \mid \theta) = \prod_{i=1}^{n} p(y_i \mid \theta) = \prod_{i=1}^{n} \theta^{y_i}(1-\theta)^{1-y_i} = \theta^{\sum_{i=1}^{n} y_i}(1-\theta)^{n-\sum_{i=1}^{n} y_i}$$

If we consider a uniform prior for θ, that is, $p(\theta) = 1, \theta \in [0, 1]$. Then the posterior is derived as

$$p(\theta \mid y_1, \ldots, y_n) \propto p(y_1, \ldots, y_n \mid \theta)p(\theta) = \theta^{\sum_{i=1}^{n} y_i}(1-\theta)^{n-\sum_{i=1}^{n} y_i}$$

In general, we consider the beta distribution

$$f(\theta) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\theta^{a-1}(1-\theta)^{b-1}$$

Note: the uniform distribution is a special case of the beta distribution when $a = b = 1$.

Now derive that the posterior distribution for θ is $\theta \mid y \sim$ beta($\sum_{i=1}^{n} y_i + 1, n - \sum_{i=1}^{n} y_i + 1$)

Given data that $n = 100$ and $\sum_{i=1}^{n} y_i = 60$, the posterior distribution is $\theta \mid y \sim$ beta$(61, 41)$.

Let's now compare the prior to the posterior. For the prior distribution beta(a, b), E$(\theta) = a/(a+b)$ and Var$(\theta) = ab/[(a+b+1)(a+b)^2]$. Write down the posterior mean and variance. Plug in the data to the posterior mean and variance.

Notice that the posterior analysis only depends on $\sum_i y_i$ (so does the likelihood function). The quantity $\sum_i y_i$ is known to be a sufficient statistic for θ in statistical inference theory. Therefore, consider a slightly different model: let $y = \sum_i y_i$, and $y \sim$ binomial(n, θ). We know that the probability mass function for a binomial random variable is as follows

$$p(y \mid \theta) = \frac{n!}{y!(n-y)!}\theta^y(1-\theta)^{n-y}$$

This is also the likelihood function of θ. The only difference between this model and the previous Bernoulli model is up a constant $\frac{n!}{y!(n-y)!}$. The two likelihood functions are proportional to each other.

(The Likelihood Principle.) If two distinct sampling plans (designs) yield proportional likelihood functions for θ, then inference about θ should be identical from these two designs. Mathematically, inference about θ should be the same for l_1 and l_2 if

$$l_1(\theta \mid \boldsymbol{x}) = C \cdot l_2(\theta \mid \boldsymbol{x}), \quad C \text{ is a constant}$$

Frequentist inference does not obey the likelihood principle.

Example: Suppose in 12 independent tosses of a coin, 9 heads and 3 tails are observed. The last observation is a tail. If θ is the probability of obtaining a head in a single toss. We want to test $H_0 : \theta = .5$ versus $H_1 : \theta > .5$.

Solution 1: This is a binomial experiment. Suppose $n = 12$ is the total number of trials and y is the number of heads in the 12 tosses. Then $y \sim binom(12, \theta)$, and the likelihood function is

$$L_1(\theta) = \binom{n}{y} \theta^y (1-\theta)^{n-y} = \binom{12}{9} \theta^9 (1-\theta)^3$$

The P-value, for the test of interest, is

$$P(y \geq 9 \mid \theta = .5) = \sum_{j=9}^{12} \binom{12}{j} \theta^j (1-\theta)^{12-j} = .075$$

Solution 2: This is a negative binomial experiment. It can be viewed as a procedure of flipping the coin until the third tail appeared. Suppose $r = 3$ is the number of tails required and y is the number of heads before the third tail. Then $y \sim nb(3, \theta)$, and the likelihood function is

$$L_2(\theta) = \binom{r+y-1}{y} \theta^y (1-\theta)^r = \binom{11}{9} \theta^9 (1-\theta)^3$$

The P-value, for the test of interest, is

$$P(y \geq 9 \mid \theta = .5) = \sum_{j=9}^{\infty} \binom{3+j-1}{j} \theta^j (1-\theta)^{12-j} = .0325$$

Notice that the likelihood function L_1 is proportional to L_2. By the likelihood principle, the inference about θ should be the same. However, the P-values are completely different. If the sharp decision rule is $\alpha = .05$, then the decisions are opposite. The reason is mentioned in the previous chapter: frequentist assume that data are a repeatable random sample, and assuming repeatability is to create a probabilistic model.

Under the Bayesian framework, the posterior is proportional to $\theta^y(1-\theta)^{n-y}$, regardless the Bernoulli model or the binomial model. Under the uniform prior

$$p(\theta \mid y) \propto \theta^y(1-\theta)^{n-y}$$

which is beta$(y+1, n-y+1)$ and the normalizing constant is

$$p(y) = \frac{\Gamma(n+2)}{\Gamma(y+1)\Gamma(n-y+1)}$$

Under the prior beta(a, b),

$$p(\theta \mid y) \propto \theta^{y+a-1}(1-\theta)^{n-y+b-1}$$

which is beta$(y+a, n-y+b)$

Definition 2. (Conjugacy) *A class $\mathcal{P}$ of prior distributions for parameter θ is called conjugate for a sampling distribution model $p(y \mid \theta)$, if*

$$p(\theta) \in \mathcal{P} \longrightarrow p(\theta \mid y) \in \mathcal{P}$$

To see how the posterior combines information from data and the prior, consider

$$\mathrm{E}(\theta \mid y) = \frac{a+y}{a+b+n} = \frac{a+b}{a+b+n} \times \frac{a}{a+b} + \frac{n}{a+b+n} \times \frac{y}{n}$$

where the prior mean is $\mathrm{E}(\theta) = a/(a+b)$ and the sample mean is y/n. As sample n increases, the posterior will be more dominated by data than the prior. The prior parameters a and b determines how informative the prior is.

Predictions. A future unobserved value is denoted by $\tilde{Y}$. Note that it is also a random variable. We are concerned about the predictive distribution $p(\tilde{Y} \mid \text{Observed Data})$. In the binomial model case, a future value is binary, and therefore, we want the probability

$$\begin{aligned}
p(\tilde{Y} = 1 \mid y_1 \ldots, y_n) &= \int p(\tilde{Y} = 1, \theta \mid y) d\theta \\
&= \int p(\tilde{Y} = 1 \mid \theta) p(\theta \mid y) d\theta \\
&= \int \theta p(\theta \mid y) d\theta \\
&= \mathrm{E}(\theta \mid y) \\
&= \frac{a+y}{a+b+n}
\end{aligned}$$

and so $p(\tilde{Y} = 0 \mid y_1 \ldots, y_n) = 1 - p(\tilde{Y} = 1 \mid y_1 \ldots, y_n) = \frac{b+n-y}{a+b+n}$.

Bayesian credible interval.

Definition 3. Bayesian Coverage. *An interval* $[l(y), u(y)]$*, based on the observed data* $Y = y$*, has* $1 - \alpha$ *Bayesian coverage for* θ*, if*

$$P[l(y) < \theta < u(y) \mid Y = y] = 1 - \alpha$$

Definition 4. Frequentist Coverage. *A random interval* $[l(Y), u(Y)]$ *has* $1 - \alpha$ *frequentist coverage for* θ*, before data are generated, if*

$$P[l(Y) < \theta < u(Y) \mid \theta] = 1 - \alpha$$

Bayesian credible intervals are intervals from the posterior distribution. We introduce two types of Bayesian intervals.

Quantile-based Interval. Use posterior quantiles to have a $1 - \alpha$ coverage:

$$P(\theta_{\alpha/2} < \theta < \theta_{1-\alpha/2} \mid y) = 1 - \alpha$$

Definition 5. (Highest Posterior Density (HPD) Region). *A $1-\alpha$ HPD region consists of $S(y) \in \Theta$ such that*

- $P(\theta \in S(y) \mid y) = 1 - \alpha$
- *If $\theta_a \in S(y)$ and $\theta_b \notin S(y)$, then $P(\theta_a \mid y) > P(\theta_b \mid y)$*

Note: the highest density region corresponds to the shortest interval.

2.2 The single-parameter Normal model

We say that a random variable follows a normal distribution $Y \sim N(\theta, \sigma^2)$ if its density function is

$$p(y \mid \theta, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{y-\theta}{\sigma}\right)^2}, \quad -\infty < y < \infty$$

When we have an i.i.d. sample, $y_1, \ldots, y_n \sim N(\theta, \sigma^2)$, the joint distribution (likelihood function) is then

$$p(y_1, \ldots, y_n \mid \theta, \sigma^2) = \prod_{i=1}^{n} p(y_i \mid \theta, \sigma^2) = (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left\{-\frac{1}{2}\sum_{i=1}^{n}\left(\frac{y-\theta}{\sigma}\right)^2\right\}$$

Assume that we want to infer θ given that σ^2 is known. It is obvious that a conjugate prior for θ is also normal: $\theta \sim N(\mu_0, \tau_0^2)$,

$$p(\theta) = \frac{1}{\sqrt{2\pi}\tau_0} e^{-\frac{1}{2}\left(\frac{\theta-\mu_0}{\tau_0}\right)^2}, \quad -\infty < \theta < \infty$$

Without derivation, now we give the posterior for θ:

$$\begin{aligned} p(\theta \mid y) &\propto p(y \mid \theta)p(\theta) \\ &\propto \exp\left\{-\frac{\sum_{i=1}^{n}(y_i-\theta)^2}{2\sigma^2}\right\}\exp\left\{-\frac{(\theta-\mu_0)^2}{2\tau_0^2}\right\} \\ &\propto \exp\left\{-\frac{(\theta-\mu_n)^2}{2\tau_n^2}\right\} \end{aligned}$$

where

$$\tau_n^2 = \frac{1}{\frac{1}{\tau_0^2}+\frac{n}{\sigma^2}}, \qquad \mu_n = \frac{\frac{1}{\tau_0^2}\mu_0 + \frac{n}{\sigma^2}\bar{y}}{\frac{1}{\tau_0^2}+\frac{n}{\sigma^2}}$$

denoted by $\theta \mid y \sim N(\mu_n, \tau_n^2)$. To see that this posterior combines information from the prior and data: $1/\tau_0^2$ is the prior precision, n/σ^2 is the sample precision, μ_0 is the prior mean, and $\bar{y}$ is the sample mean.

Prediction. Consider predicting a future value $\tilde{y}$. The predictive distribution is $\tilde{y} \mid (\sigma^2, y) \sim N(\mu_n, \tau_n^2 + \sigma^2)$.

2.3 The Poisson model

Let random variable $Y = 0, 1, 2, \ldots$ follow a Poisson distribution Poi(λ). The Poisson distribution is often used to model a count of rare event. Mathematically, it is the limiting distribution of binomial(n, p), when $n \to +\infty$, $p \to 0$, and $np \to \lambda$. Practically, it means a rare event ($p \to 0$) for a large number of trials ($n \to +\infty$), such as infection count in a city, number of accidents at a traffic light. Recall that the mean for a binomial is np, so λ is the mean of the Poisson. The probability mass function of Poisson is given by (use notation θ instead of λ)

$$P(Y = y \mid \theta) = \frac{\theta^y e^{-\theta}}{y!}$$

If we observe a sample $y_1, \ldots, y_n \sim$ Poi(θ) i.i.d. the likelihood function can be written as

$$p(y_1, \ldots, y_n \mid \theta) = \prod_{i=1}^{n} p(y_i \mid \theta) = \prod_{i=1}^{n} \frac{\theta^{y_i} e^{-\theta}}{y_i!} = \left(\frac{1}{\prod_i y_i!} \right) \theta^{\sum_i y_i} e^{-n\theta}$$

Note, here $\sum_i y_i$ is a sufficient statistic for θ, and the frequentist estimator for θ is $\sum_i y_i / n$.

Conjugate prior. If we observe the likelihood function in terms of θ, it is not difficult to find that the conjugate prior for θ is the gamma distribution Gamma(a, b).

$$p(\theta) = \frac{b^a}{\Gamma(a)}\theta^{a-1}e^{-b\theta}, \theta, a, b > 0$$

with

$$\mathrm{E}(\theta) = \frac{a}{b}, \quad \mathrm{Var}(\theta) = \frac{a}{b^2}$$

Derive the posterior using Bayes rule

$$p(\theta \mid y_1, \ldots, y_n) \propto p(y_1, \ldots, y_n \mid \theta)p(\theta) \propto \theta^{a+\sum_i y_i - 1}e^{-(b+n)\theta}$$

which is Gamma$(a + \sum_i y_i, b + n)$.

The normalizing constant for this gamma distribution is $\frac{(b+n)^{a+\sum_i y_i}}{\Gamma(a+\sum_i y_i)}$.

To see that the posterior combines information from data and the prior,

$$\mathrm{E}(\theta \mid y) = \frac{a + \sum_i y_i}{b+n} = \frac{b}{b+n} \times \frac{a}{b} + \frac{n}{b+n} \times \frac{\sum_i y_i}{n}$$

where a/b is the prior mean and $\sum y_i/n$ is the sample mean. As $n \to \infty$, $\mathrm{E}(\theta \mid y) \to \bar{y}$ and $\mathrm{Var}(\theta \mid y) \to \bar{y}/n$, which is exactly the same as frequentist solution.

Derive the predictive distribution for a future value $\tilde{Y}$: $\tilde{Y} \mid y \sim$ Neg-Bin$(a + \sum_i y_i, \frac{b+n}{b+n+1})$.

3 Monte Carlo

3.1 Monte Carlo approximation

In the Poisson model, let's say we have posterior distributions $\theta_1 \mid y_1 \sim \text{Gamma}(219, 112)$ and $\theta_2 \mid y_2 \sim \text{Gamma}(68, 45)$. If our goal is to find the probability $P(\theta_1 > \theta_2 \mid y_1, y_2)$, we certainly could compute this analytically

$$P(\theta_1 > \theta_2 \mid y_1, y_2) = \int_0^\infty \int_0^{\theta_1} p(\theta_1 \mid y_1) p(\theta_2 \mid y_2) d\theta_2 d\theta_1$$

This isn't always trivial to do. Can we have a universal solution for any type of posterior inference?

Theorem 1. (Law of Large Numbers) *Let $X_1, \ldots, X_n$ be i.i.d. samples from distribution F with mean μ*

$$\bar{X} \to \mu = E(X) \quad \textit{as} \quad n \to \infty$$

which also implies that $\frac{1}{n}\sum_i g(X_i) \to \mathrm{E}[g(X)] = \int g(x) f(x) dx$, where $f(x)$ is the density function of X and $g()$ is an arbitrary function.

The law of large numbers provides a way to use i.i.d. samples to approximate an expectation, i.e., an integral. Most inference targets can be formulated as an integral, and thus can be approximated numerically. This is known as the **Monte Carlo Approximation** of an integral. Revisit the integral at the beginning of this chapter. How can that probability be approximated by the Monte Carlo method?

In fact, if we can draw i.i.d. samples $\theta^{(1)}, \ldots, \theta^{(S)}$ from the posterior distribution $p(\theta \mid y)$, most posterior inference can be done by computation.

- $\frac{1}{S}\sum_{s=1}^{S}\theta^{(s)} \to \mathrm{E}(\theta \mid y)$
- $\frac{\sum_{s=1}^{S}(\theta^{(s)}-\bar{\theta})^2}{S-1} \to \mathrm{Var}(\theta \mid y)$
- $\frac{\#\{\theta^{(s)}<C\}}{S} \to P(\theta < C \mid y)$
- The empirical distribution (such as histogram) of $\theta^{(1)}, \ldots, \theta^{(S)}$ approximates $p(\theta \mid y)$
- The median of $\theta^{(1)}, \ldots, \theta^{(S)}$ approximates the median of $p(\theta \mid y)$
- The α-percentile approximates $\theta_\alpha \mid y$

3.2 Posterior inference

Suppose we are interested in $g(\theta) \mid y$ in a binomial model. For instance, we are interested in the log odds ratio

$$g(\theta) = \log \frac{\theta}{1-\theta} = \gamma$$

Now we know $p(\theta \mid y)$, how do we infer $\gamma \mid y$?

Draw samples $\theta^{(1)}, \ldots, \theta^{(S)} \sim p(\theta \mid y)$. Compute $\gamma^{(1)} = g(\theta^{(1)}), \gamma^{(2)} = g(\theta^{(2)}), \ldots, \gamma^{(S)} = g(\theta^{(S)})$. The sample $\gamma^{(1)}, \ldots, \gamma^{(S)}$ represents exact i.i.d. draws from $p(\gamma \mid y)$.

3.3 Predictive inference

The predictive distribution is obtained by integration as well: $p(\tilde{y} \mid y) = \int p(\tilde{y} \mid \theta) p(\theta \mid y) d\theta$. Therefore, the predictive inference can be done by Monte Carlo methods as well.

Recall that in the Poisson model, $\tilde{y} \mid \theta \sim \text{Poi}(\theta)$ and $\theta \mid y \sim$ Gamma, and then the predictive distribution $\tilde{y} \mid y$ follows a negative-binomial distribution. Using Monte Carlo methods, we can sample sequentially from gamma and Poisson to obtain i.i.d. samples for $\tilde{y}$.

1. Sample $\theta^{(s)} \sim p(\theta \mid y)$
2. Sample $\tilde{y}^{(s)} \sim p(y \mid \theta^{(s)})$
3. Repeat the process for S times.

At the end, we obtain samples $\{\tilde{y}^1, \ldots, \tilde{y}^S\}$ which can be used for Monte Carlo approximations for $p(\tilde{y} \mid y)$.

4 Multiple-parameter models

4.1 The Normal models

Recall that a random variable follows a normal distribution $Y \sim N(\theta, \sigma^2)$ if its density function is

$$p(y \mid \theta, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{y-\theta}{\sigma}\right)^2}, \quad -\infty < y < \infty$$

When we have an i.i.d. sample, $y_1, \ldots, y_n \sim N(\theta, \sigma^2)$, the joint distribution (likelihood function) is then

$$p(y_1, \ldots, y_n \mid \theta, \sigma^2) = \prod_{i=1}^{n} p(y_i \mid \theta, \sigma^2) = (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left\{-\frac{1}{2}\sum_{i=1}^{n}\left(\frac{y-\theta}{\sigma}\right)^2\right\}$$

Now $Y \sim N(\theta, \sigma^2)$ with both parameters unknown. The conjugate prior for (θ, σ^2) can only be assumed in a conditional setting:

$$p(\theta, \sigma^2) = p(\theta \mid \sigma^2)p(\sigma^2)$$

where $p(\sigma^2) \sim$ Inverse-gamma$(\nu_0/2, \nu_0\sigma_0^2/2)$, equivalently $1/\sigma^2 \sim$ Gamma$(\nu_0/2, \nu_0\sigma_0^2/2)$; and $p(\theta \mid \sigma^2) \sim N(\mu_0, \sigma^2/\kappa_0)$.

Side note: an inverse-gamma random variable is the reciprocal of a gamma random variable. It has two parameterizations: IG(a, b) or $IG(\nu_0/2, \nu_0\sigma_0^2/2)$ which is related to a scaled-$\chi^2(\nu_0, \sigma_0^2)$ distribution.

The conjugate prior for (θ, σ^2) is then called a normal-inverse-gamma prior with four hyper-parameters: N-IG$(\mu_0, \kappa_0, \sigma_0^2, \nu_0)$. This is the only way to build a conjugacy in the normal model.

With this prior, we can derive the posterior distribution for (θ, σ^2), also a N-IG distribution: $(\theta, \sigma^2) \mid y \sim$ N-IG$(\mu_n, \kappa_n, \sigma_n^2, \nu_n)$, which implies that the posterior distribution can be decomposed conditionally

$$p(\theta, \sigma^2 \mid y) = p(\theta \mid \sigma^2, y)p(\sigma^2 \mid y)$$

where $\sigma^2 \mid y \sim$ IG$(\nu_n/2, \nu_n\sigma_n^2/2)$ with

$$\begin{aligned} \nu_n &= \nu_0 + n \\ \sigma_n^2 &= \frac{1}{\nu_n}\left[\nu_0\sigma_0^2 + (n-1)s^2 + \frac{\kappa_0 n}{\kappa_0 + n}(\bar{y} - \mu_0)^2\right] \end{aligned}$$

and $\theta \mid \sigma^2, y \sim N(\mu_n, \sigma^2/\kappa_n)$ with

$$\begin{aligned} \kappa_n &= \kappa_0 + n \\ \mu_n &= \frac{\kappa_0\mu_0 + n\bar{y}}{\kappa_0 + n} \end{aligned}$$

To perform Monte Carlo approximations in posterior inference, we need to draw i.i.d. samples from the posterior distribution N-IG$(\mu_n, \kappa_n, \sigma_n^2, \nu_n)$. Due to its conditional nature, this sampling can be done sequentially, that is, to draw σ^2 samples from IG$(\nu_n/2, \nu_n\sigma_n^2/2)$ first, and then given those σ^2 samples, draw θ samples from $N(\mu_n, \sigma^2/\kappa_n)$.

The procedure is done as follows, for $s = 1, \ldots, S$,

1. Draw $\sigma^2_{(s)} \sim$ IG$(\nu_n/2, \nu_n\sigma_n^2/2)$
2. Draw $\theta_{(s)} \sim N(\mu_n, \sigma^2_{(s)}/\kappa_n)$

This ensures that $\{\sigma^2_{(1)}, \ldots, \sigma^2_{(S)}\}$ are exact i.i.d. samples from its marginal posterior distribution $p(\sigma^2 \mid y)$, and $\{\theta_{(1)}, \ldots, \theta_{(S)}\}$ are exact i.i.d. samples from its marginal posterior distribution $p(\theta \mid y)$ (due to Monte Carlo approximation).

Side note: what is the difference between the conditional posterior $p(\theta \mid \sigma^2, y)$ and the marginal posterior $p(\theta \mid y)$?

In Bayesian analysis, we often want to choose "non-informative" priors to minimize the impact of the prior distribution, so that our analysis is as "objective" as possible. In the normal model, to achieve this, we can let $\kappa_0 \to 0$ and $\nu_0 \to 0$, which ensures that

$$\mu_n \to \bar{y}, \qquad \sigma_n^2 \to \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2$$

which is consistent with the frequentist solution.

However, such a choice of κ_0 and ν_0 will make the prior distribution invalid density functions:

$$\begin{aligned} p(\theta) &\propto 1, \quad -\infty < \theta < \infty \\ p(\sigma^2) &\propto \frac{1}{\sigma^2}, \quad \sigma^2 > 0 \end{aligned}$$

Integration of either one is infinity: $\int_{-\infty}^{\infty} p(\theta)d\theta = \infty$ and $\int_0^{\infty} \frac{1}{\sigma^2} d\sigma^2 = \infty$. They are no longer valid normal or IG distributions, but they are the limiting case of normal and IG. We call these prior distributions the improper priors.

4.2 The Multinomial models

The Multinomial model is defined as

$$(y_1, \ldots, y_k) \sim \text{multinomial}(n, \theta_1, \ldots, \theta_k)$$

subject to that $\sum_{i=1}^{k} y_i = n$ and $\sum_{i=1}^{k} \theta_i = 1$.

The binomial model is a special case of the multinomial model when $k = 2$.

A frequentist estimator for θ is $\hat{\theta}_i = y_i/n$.

The conjugate prior for the multinomial model is the Dirichlet distribution:

$\theta = (\theta_1, \ldots, \theta_k) \sim \text{Dir}(\alpha_1, \ldots, \alpha_k)$, with its density function as follows

$$p(\theta) = \frac{\Gamma(\sum \alpha_i)}{\Gamma(\alpha_1) \cdots \Gamma(\alpha_k)} \theta_1^{\alpha_1 - 1} \cdots \theta_k^{\alpha_k - 1}$$

We know that the conjugate prior for the binomial model is beta distribution. Show that the beta distribution is a special case of the Dirichlet distribution.

Properties of the Dirichlet distribution. Mean, variance, covariance.

Due to the conjugacy, the posterior distribution for θ is also Dirichlet.

$$p(\theta \mid y) \propto \theta_1^{y_1 + \alpha_1 - 1} \theta_2^{y_2 + \alpha_2 - 1} \cdots \theta_k^{y_k + \alpha_k - 1}$$

or $\theta \mid y \sim \text{Dir}(y_1 + \alpha_1, \ldots, y_k + \alpha_k)$. Posterior inference follows.

Example: In the development of drugs, acute toxicity tests or bioassay experience are commonly performed on animals. The animals' responses are characterized by a dichotomous outcome: alive or dead, tumor or no tumor. The data are in the form (x_i, n_i, y_i).

Table 1: Bioassay data from Racine et al. (1986)

Dose, x_i (log g/ml)	Number of animals. n_i	Number of deaths, y_i
-0.86	5	0
-0.30	5	1
-0.05	5	3
0.73	5	5

Let p_i be the probability of death at i-th level, then $y_i \sim \text{binom}(n_i, p_i)$. To link the death probability p_i with the explanatory variable x_i, the logistic regression is commonly used.

$$\text{logit}(p_i) = \log \frac{p_i}{1-p_i} = \beta_0 + \beta_1 x_i$$

A simple analysis would treat y_i independent of each other. Then the likelihood function is

$$\begin{aligned} p(y \mid \beta_0, \beta_1, n_i, x_i) &= \prod_{i=1}^{k} p(y_i \mid \beta_0, \beta_1) = \prod_{i=1}^{k} \binom{n_i}{y_i} p_i^{y_i} (1-p_i)^{n_i - y_i} \\ &= \prod_{i=1}^{k} \binom{n_i}{y_i} \left(\frac{1}{1+e^{-\beta_0-\beta_1 x_i}} \right)^{y_i} \left(\frac{e^{-\beta_0-\beta_1 x_i}}{1+e^{-\beta_0-\beta_1 x_i}} \right)^{n_i - y_i} \end{aligned}$$

The conjugate prior for (β_0, β_1) is not available.

Posterior inference with informative prior:

A noninformative prior can be used: $p(\beta_0, \beta_1) \propto 1$. Derive the posterior in this case.

Posterior inference using Monte Carlo methods needs i.i.d. samples. However, drawing samples from such a posterior distribution is not obvious. In future chapters, we will introduce the Markov Chain Monte Carlo techniques. For now, suppose that we can obtain samples from the posterior. A parameter of interest in bioassay studies is the $LD50$ – the dose level at which the death probability is 50%.

$$\text{logit}(p_i) = \log(.5/(1-.5)) = 0 \rightarrow \beta_0 + \beta_1 x_i = 0 \rightarrow x_i = -\frac{\beta_0}{\beta_1}$$

Hence the $LD50 = -\beta_0/\beta_1$.

5 Markov Chain Monte Carlo and posterior approximation

5.1 Gibbs sampling

We start from an example of the normal model: $y \sim N(\theta, \sigma^2)$. We know that the conjugate prior is N-IG, which leads to an N-IG posterior that can be sequentially sampled. Now let us consider a semi-conjugate prior, where θ and σ^2 are independent: $p(\theta, \sigma^2) = p(\theta)p(\sigma^2)$. We may use $N(\mu_0, \tau_0^2)$ and $\text{IG}(\nu_0/2, \nu_0\sigma_0^2/2)$ as priors. While this prior is not conjugate, the posterior distribution will not be normal and inverse-gamma. Instead, it will be some unrecognizable distribution that needs to be approximated.

Let us first derive the conditional posterior when the other parameter is given (known).

$$p(\sigma^2 \mid \theta, y) \propto p(y \mid \theta, \sigma^2)p(\sigma^2)$$

We can derive that this posterior is $\text{IG}(\nu_n/2, \nu_n\sigma_n^2(\theta)/2)$, where

$$\begin{aligned} \nu_n &= \nu_0 + n \\ \sigma_n^2(\theta) &= \frac{1}{\nu_n}\left[\nu_0\sigma_0^2 + \sum_{i=1}^{n}(y_i - \theta)^2\right] \end{aligned}$$

So that given a fixed θ, it is easy to sample σ^2.

Similarly, given σ^2, $p(\theta \mid \sigma^2, y) \propto p(y \mid \theta, \sigma^2)p(\theta)$, which is $N(\mu_n(\sigma^2), \tau_n(\sigma^2))$, where

$$\begin{aligned} \tau_n^2(\sigma^2) &= \frac{1}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}} \\ \mu_n(\sigma^2) &= \frac{\frac{\mu_0}{\tau_0^2} + \frac{n\bar{y}}{\sigma^2}}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}} \end{aligned}$$

which is easy to sample from as well.

Now we can easily sample from each of the conditional posterior distributions, given the other parameter, but don't know how to sample from their marginal posterior distributions $p(\theta \mid y)$ and $p(\sigma^2 \mid y)$.

The answer is to combine them, and that is called the Gibbs sampling. Gibbs sampling is an algorithm that iteratively sample from these conditional posterior distributions, known as the full conditional distributions. In our case, choose an initial value $\theta_{(0)}$ or $\sigma^2_{(0)}$ and do the following

1. Draw $\theta_{(s)}$ from $N(\mu_n(\sigma^2_{(s-1)}), \tau_n(\sigma^2_{(s-1)}))$

2. Draw $\sigma_{(s)}$ from $\text{IG}(\nu_n/2, \nu_n\sigma_n^2(\theta_{(s)})/2)$

for $s = 1, \ldots, S$. The order of step 1 and 2 does not matter, but parameters must be updated iteratively. It is theoretically guaranteed that the draws $\{\theta_{(1)}, \ldots, \theta_{(S)}\}$ and $\{\sigma^2_{(1)}, \ldots, \sigma^2_{(S)}\}$ can be used to approximate posterior inference just as i.i.d. draws when $S \to \infty$, even that the chains are not i.i.d. samples.

We essentially sample from full conditional distributions (which are easy to sample from) to approximate the desired marginal posteriors.

The general Gibbs sampling works as follows. Suppose that we have a vector of parameters $\phi = (\phi_1, \ldots, \phi_p)$, and we want to sample from the distribution $p(\phi_1, \ldots, \phi_p)$. First, we need to derive the full conditional distributions, $p(\phi_1 \mid \phi_2, \ldots, \phi_p)$, $p(\phi_2 \mid \phi_1, \phi_3, \ldots, \phi_p)$, until $p(\phi_p \mid \phi_1, \ldots, \phi_{p-1})$. In general, they can be denoted by $p(\phi_i \mid \text{others})$. Then, iteratively sample the following

1. $\phi_1^{(s)} \sim p(\phi_1 \mid \phi_2^{(s-1)}, \ldots, \phi_p^{(s-1)})$
2. $\phi_2^{(s)} \sim p(\phi_2 \mid \phi_1^{(s)}, \phi_3^{(s-1)}, \ldots, \phi_p^{(s-1)})$

3. until
4. $\phi_p^{(s)} \sim p(\phi_p \mid \phi_1^{(s)}, \ldots, \phi_{p-1}^{(s)})$

The generated chain of samples from $s = 1$ to S is called a Markov chain, which possesses the Markov property:

$$\phi^{(s)} \mid (\phi^{(s-1)}, \phi^{(s-2)}, \cdots, \phi^{(1)}) = \phi^{(s)} \mid \phi^{(s-1)}$$

5.2 MCMC

The Gibbs sampling algorithm generates a Markov chain. Not every Markov chain can converge to a stationary distribution (theory of stochastic processes). Under certain conditions (irreducible, positive recurrent, time reversible), as $S \to \infty$,

$$P(\phi^{(s)} \in \mathcal{A}) \to \int_{\mathcal{A}} p(\phi) d\phi$$

In other words, the sampling distribution of the Markov chain $\phi^{(s)}$ approaches the true target distribution $p(\phi)$ in the long run, no matter what initial value $\phi^{(0)}$ you have specified.

Note that the Markov chain is not i.i.d. samples. They are sequentially dependent and so represent a time series with non-zero autocorrelation. The Law of Large Numbers requires an i.i.d. sample, so the Monte Carlo approximation cannot be readily used. Here, the theoretical guarantee is the ergodic theorem, which basically says that, given a convergent Markov chain, we still

have a Monte Carlo approximation

$$\frac{1}{S}\sum_{s=1}^{S} g(\phi^{(s)}) \to \mathrm{E}[g(\phi)] = \int g(\phi)p(\phi)d\phi$$

as $S \to \infty$. Theorem ensures the use of Markov chain samples to do a Monte Carlo approximation, and this is referred to as the Markov Chain Monte Carlo (MCMC).

Note: we need to distinguish the terms (parameter) estimation and (posterior) approximation, although they appear to be similar. When you have a posterior distribution, using the posterior distribution to estimate θ, the variance, the quantiles, etc., is parameter estimation and inference. They need not to be approximated, for instance, in the binomial model, θ can be estimated by a closed-form expression. On the other hand, when the posterior distribution is non-trivial, estimation cannot be done analytically. We use MCMC to approximate these estimation (or inference), without needing the closed-form expressions.

Introduction to MCMC diagnostics. Suppose our targeted distribution is the joint distribution of (δ, θ), where $\delta = 1, 2,$ or 3 with corresponding proba-

bilities 0.45, 0.10, and 0.45, respectively, and $(\theta|\delta = j) \sim N(\mu_j, 1/3)$, where $(\mu_1, \mu_2, \mu_3) = (-3, 0, 3)$. Although we can obtain independent MC samples from the joint distribution by first sampling from $p(\delta)$ and subsequently sampling from $p(\theta|\delta)$, we can also construct a Gibbs sampler using the full condition distributions. Specifically,

$$p(\delta = j|\theta) = \frac{p(\delta = j) \times p(\theta|\delta = j)}{\sum_{j'=1}^{3} p(\delta = j') \times p(\theta|\delta = j')}.$$

The chain represents a trajectory of a particle ϕ moving around its parameter space. The amount of time ϕ spends in a given set $\mathcal{A}$ should be proportional to the probability $P(\phi \in \mathcal{A}) = \int_{\mathcal{A}} p(\phi) d\phi$.

In the mixture normal case, we have roughly three regions $P(\mathcal{A}_2) < P(\mathcal{A}_1) \approx P(\mathcal{A}_3)$. The particle should spend the correct amount of time in each region, respectively.

In the scenario when the particle is trapped in a region, it needs a long enough chain to move out of the region, and spend enough time in other regions as well. We need a large enough S so that (1) the particle has a chance to move out of $\mathcal{A}_2$ and enter a higher probability region, and (2) the particle moves freely between $\mathcal{A}_1$ and $\mathcal{A}_3$. This is not necessarily easy to achieve.

Besides that, the chain must converge to the stationary distribution, and we also hope the chain to move appropriately in the parameter space, which is called the speed of mixing. A chain can mix well or mix poorly, even though theoretically it is guaranteed to converge when S is large (but when?). A poorly mixed chain needs a much large S.

The speed of mixing can be reflected by the autocorrelation of the time series. A high autocorrelation implies a poor mixing. We can study this by calculating the MCMC approximation error vs. the i.i.d. MC error.

$$\text{Var}_{MCMC}(\bar{\phi}) = \text{Var}_{MC}(\bar{\phi}) + \frac{1}{S^2}\sum_{s \neq t} \text{E}[(\phi^{(s)} - \phi_0)(\phi^{(t)} - \phi_0)]$$

where the second term determines the autocorrelation. A high autocorrelation increases the MCMC approximation error.

Another useful concept is the effective sample size, computed by

$$S_{eff} = \frac{\text{Var}[\phi]}{\text{Var}_{MCMC}[\bar{\phi}]}$$

which represents the number of i.i.d. sample the S-sized chain is equivalent to. In the mixture model example, although $S = 10,000$, $S_{eff} = 18.42$.

5.3 The Laplace approximation

When the posterior distribution is not a commonly used distribution, we have two approaches to the posterior inference: 1. Draw samples and use Monte Carlo approximations; 2. Approximate the posterior with a commonly used distribution. For the second approach, a widely acceptable choice is the normal distribution, known as the Laplace approximation.

Consider the logarithm of the joint

$$h(\theta, y) = \log(p(y \mid \theta)p(\theta))$$

Denote this by $h(\theta)$ when y is given (actually the log-posterior). Apply a second-order Taylor expansion at its mode $\hat{\theta}$

$$h(\theta) \approx h(\hat{\theta}) + (\theta - \hat{\theta})' h''(\hat{\theta})(\theta - \hat{\theta})/2$$

where h'' is the Hessian matrix of h. Since this is the logarithm of the target posterior, let's now transform it exponentially back:

$$e^{h(\hat{\theta})} e^{(\theta-\hat{\theta})' h''(\hat{\theta})(\theta-\hat{\theta})/2}$$

which implies a Gaussian (normal) density with mean vector $\hat{\theta}$ and covariance matrix $[-h''(\hat{\theta})]^{-1}$. This is the Laplace approximation: using a Gaussian distribution $N(\hat{\theta}, [-h''(\hat{\theta})]^{-1})$ to approximate the unknown posterior $p(\theta \mid y)$. This method requires an optimization of $h(\theta)$ to obtain its mode, typically done by Newton's method or other related techniques.

Example: logistic regression.

5.4 Rejection sampling

Suppose that we want to sample from a posterior density $p(\theta \mid y)$. We specify a proposal density (also called an envelope density) $g(\theta)$, which satisfy the following conditions:

1. $g(\theta)$ is easy to sample from
2. Support of $g(\theta)$ is at least the same as support of $p(\theta \mid y)$
3. For all θ, $p(\theta \mid y) < Cg(\theta)$, for some finite C

The Rejection-Acceptance Algorithm is as follows.

1. Sample $\theta \sim g(\theta)$ and $u \sim \text{unif}(0, 1)$
2. If $u \leq \frac{p(\theta|y)}{Cg(\theta)}$, accept θ as a random draw from the target distribution. Otherwise, reject this draw

3. Repeat until one has collected a sufficient number of samples

A few notes:

- There is a mathematical proof that guarantees the draws are exact
- The efficiency is measured by the rejection rate or, equivalently, the acceptance rate
- A convenient choice of $g(\theta)$ can always be a uniform distribution, but highly inefficient
- To have an efficient rejection sampling, one needs $g(\theta)$ to be as close to $p(\theta \mid y)$ as possible. A good practice is to let $g(\theta)$ be the Laplace approximation of $p(\theta \mid y)$

5.5 Importance sampling

Importance sampling is used to evaluate an integral. We have previously introduced Monte Carlo methods for the same purpose. The importance sampling relies on the following derivation

$$\begin{aligned}
\mathrm{E}(h(\theta) \mid y) &= \int h(\theta) p(\theta \mid y) d\theta \\
&= \int h(\theta) \frac{g(\theta)}{g(\theta)} p(\theta \mid y) d\theta \\
&= \int h(\theta) \frac{p(\theta \mid y)}{g(\theta)} g(\theta) d\theta = \mathrm{E}(h(\theta) w(\theta \mid y))
\end{aligned}$$

where $w(\theta \mid y) = \frac{p(\theta|y)}{g(\theta)}$ is known as the importance weight. The target integral now can be approximated by a weighted function under the measure of $g(\theta)$, which is easy to sample from. To use the Monte Carlo methods, sample $\theta^{(s)} \sim g(\theta), s = 1, \ldots, S$, and

$$\mathrm{E}(h(\theta) \mid y) \approx \sum_{s=1}^{S} h(\theta^{(s)}) w(\theta^{(s)})$$

Sampling Importance Resampling. The algorithm can be improved by a resampling step. Normalize the weights to probabilities $p^j = \frac{w^j}{\sum w^j}$. Resample (with replacement) S samples from $\theta^{(s)} \sim g(\theta), s = 1, \ldots, S$ according to probabilities p^j.

Example. $X \sim N(0, 1)$, find $P(X > 4)$ through simulation.

5.6 Metropolis-Hastings algorithm

The MH algorithm constructs a Markov chain and hence is an MCMC technique. It is similar to the rejection sampling but less restrictive on the proposal density. Suppose that we want to sample from $p(\theta \mid y)$. For the sake of simplicity, we drop the given y notation and denote the target by $p(\theta)$. The MH algorithm needs a proposal (jumping) density in the form $g(\theta^* \mid \theta^{t-1})$, which depends on the previous draw θ^{t-1}. The algorithm is as follows.

1. Draw a candidate value $\theta^* \sim g(\theta^* \mid \theta^{t-1})$
2. Compute the ratio
$$r = \frac{p(\theta^* \mid y)g(\theta^{t-1} \mid \theta^*)}{p(\theta^{t-1} \mid y)g(\theta^* \mid \theta^{t-1})}$$
3. Compute the acceptance probability $\alpha = \min\{1, r\}$
4. Let $\theta^t = \theta^*$ with probability α and let $\theta^t = \theta^{t-1}$ with probability $1 - \alpha$

In practice, the last two steps can be achieved by simulating $u \sim \text{unif}(0, 1)$, and if $\alpha \geq u$, then $\theta^t = \theta^*$, otherwise $\theta^t = \theta^{t-1}$. Also, because likelihood functions are usually evaluated on the logarithmic scale, we usually calculate $\log r$ and compare with $\log u$.

The MH algorithm produced a Markov chain $\theta^{(1)}, \ldots, \theta^{(S)}$ that is guaranteed to converge to the target posterior distribution when $S \to \infty$, regardless of the initial value and choice of proposal. However, in practice we do not have infinite computing capability, and so the speed of convergence (mixing of the chain) is important, which may depend on the initial value, the proposal density choice, and the algorithm itself.

For a special case where the proposal distribution is symmetric, i.e., $g(\theta^{t-1}|\theta^*) = g(\theta^*|\theta^{t-1})$, the ratio becomes $r = \frac{p(\theta^*|y)}{p(\theta^{t-1}|y)}$ and the algorithm is the original Metropolis algorithm. Symmetric proposal distributions, such as normal distributions, explore the sample space through a random-walk Markov Chain, which can be inefficient in high dimensions. Nonrandom walk Markov Chains that depend on gradient information usually result in an asymmetric proposal distribution.

For a random-walk jumping distribution, an optimal choice is $g(\theta^*|\theta^{t-1}) = N(\theta^*|\theta^{t-1}, c^2\Sigma)$, where $c \approx 2.4/\sqrt{d}$ and Σ is the covariance matrix of the posterior distribution (can be approximated through Laplace).

Other sampling techniques, such as slice sampling and reverse jump sampling, can be incorporated into the MH algorithm.

Gibbs sampling is a special case of MH algorithm with $\alpha = 1$.

Example: $p(x) \propto \exp(x + 4x^2 - 2x^3 - 4x^4)$.

6 Bayesian linear regression model

6.1 Conditional model

The quantity of interest, Y, is called the response or outcome variable. The variables $\boldsymbol{X} = (X_1, \cdots, X_p)^\top$ are explanatory variables. In linear regression, we assume Y is continuous, and X can be either continuous or discrete. The most widely used model is the normal linear model, where $Y|\boldsymbol{X} \sim N(\boldsymbol{X}^\top \boldsymbol{\beta}, \sigma^2)$. The parameter vector is then $\boldsymbol{\theta} = (\boldsymbol{\beta}, \sigma^2)$.

A full Bayesian model should include the joint distribution of $(Y, \boldsymbol{X})$, and thus involves a joint likelihood, $p(\boldsymbol{X}, y|\boldsymbol{\theta}, \psi)$, as well as a joint prior $p(\boldsymbol{\theta}, \psi)$.

We usually "ignore" the information supplied by X about $(\boldsymbol{\theta}, \psi)$ by assuming it provides no information about the conditional distribution of Y given $\boldsymbol{X}$, which indicates $p(\boldsymbol{\theta}, \psi) = p(\boldsymbol{\theta})p(\psi)$

6.2 Bayesian estimation

Given a sample $(y_1, x_1), \ldots, (y_n, x_n)$, the vector of observations $\boldsymbol{y} = (y_1, \cdots, y_n)^\top$ has a multivariate normal distribution: $\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{\beta}, \sigma^2 \sim \text{MVN}(\boldsymbol{X}\boldsymbol{\beta}, \sigma^2 \boldsymbol{I})$.

The least square estimation is to minimize

$$\text{SSR}(\boldsymbol{\beta}) = \sum_{i=1}^{n} (y_i - \boldsymbol{\beta}' \boldsymbol{x}_i)^2 = (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})$$

Solving this optimization yields $\hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$ which is known as the OLS, given that $(\boldsymbol{X}'\boldsymbol{X})^{-1}$ exists.

The likelihood function is

$$p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{\beta}, \sigma^2) \propto \exp\left\{-\frac{1}{2\sigma^2}\text{SSR}(\boldsymbol{\beta})\right\} \propto \exp\left\{-\frac{1}{2\sigma^2}[\boldsymbol{y}'\boldsymbol{y} - 2\boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{y} + \boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta}]\right\}$$

which suggests that the conjugate prior for $\boldsymbol{\beta}$ is multivariate normal when σ^2 is given: $\boldsymbol{\beta} \sim \text{MVN}(\boldsymbol{\beta}_0, \boldsymbol{\Sigma}_0)$.

This leads to posterior given σ^2: $\boldsymbol{\beta} \mid \boldsymbol{y}, \boldsymbol{X}, \sigma^2 \sim \text{MVN}(\boldsymbol{\beta}_n, \boldsymbol{\Sigma}_n)$, where

$$\begin{aligned}
\boldsymbol{\beta}_n &= (\boldsymbol{\Sigma}_0^{-1} + \frac{1}{\sigma^2}\boldsymbol{X}'\boldsymbol{X})^{-1}(\boldsymbol{\Sigma}_0^{-1}\boldsymbol{\beta}_0 + \frac{1}{\sigma^2}\boldsymbol{X}'\boldsymbol{y}) \\
\boldsymbol{\Sigma}_n &= (\boldsymbol{\Sigma}_0^{-1} + \frac{1}{\sigma^2}\boldsymbol{X}'\boldsymbol{X})^{-1}
\end{aligned}$$

When $\boldsymbol{\Sigma}_0^{-1}$ is small (in terms of a matrix metric), then $\boldsymbol{\beta}_n \to (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$, the OLS.

The conjugate prior for σ^2 given $\boldsymbol{\beta}$ is inverse-gamma. Let $\gamma = 1/\sigma^2$, $\gamma \sim \text{gamma}(\nu_0/2, \nu_0\sigma_0^2/2)$, which leads to the conditional posterior

$$p(\gamma \mid \boldsymbol{y}, \boldsymbol{X}, \boldsymbol{\beta}) \propto \gamma^{\frac{\nu_0+n}{2}-1} \exp\left\{-\frac{\gamma}{2}[\nu_0\sigma_0^2 + \text{SSR}(\boldsymbol{\beta})]\right\}$$

which is $\text{gamma}(\frac{1}{2}(\nu_0+n), \frac{1}{2}[\nu_0\sigma_0^2 + \text{SSR}(\boldsymbol{\beta})])$

The above results give us the full conditional distributions, and hence a Gibbs sampling can be done for posterior inference.

6.3 Prior distributions

Default and Weakly Informative Priors. Alternatively, recall that for the normal model, we may have the conjugate prior for both the mean and variance: normal-inverse-gamma. Here, we need $p(\boldsymbol{\beta} \mid \sigma^2)$ and $p(\sigma^2)$. Kass & Wasserman (1995) suggests the unit informative prior due to the fact that for the OLS $\hat{\boldsymbol{\beta}}^{OLS} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$, we have

$$\mathrm{E}(\hat{\boldsymbol{\beta}}^{OLS}) = \boldsymbol{\beta}, \qquad \mathrm{Var}(\hat{\boldsymbol{\beta}}^{OLS}) = \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}$$

The unit informative prior is

$$\boldsymbol{\beta} \mid \sigma^2 \sim N(\boldsymbol{\beta}_0 = \hat{\boldsymbol{\beta}}^{OLS}, \boldsymbol{\Sigma}_0 = n\sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1})$$

Zellner's g-prior. Let us first consider the invariance rule: estimation should be invariant to the unit change of $\boldsymbol{X}$. Mathematically, suppose that $\tilde{\boldsymbol{X}} = \boldsymbol{X}\boldsymbol{H}$ and $\boldsymbol{X}\boldsymbol{\beta} = \tilde{\boldsymbol{X}}\tilde{\boldsymbol{\beta}}$, the posterior distribution of $\boldsymbol{\beta}$ and $\boldsymbol{H}\tilde{\boldsymbol{\beta}}$ should be the same.

The unit informative prior is not invariant. To satisfy the invariance condition, one must have $\boldsymbol{\beta}_0 = \mathbf{0}$ and $\boldsymbol{\Sigma}_0 = k(\boldsymbol{X}'\boldsymbol{X})^{-1}$ for some k.

Let $k = g\sigma^2$, we have Zellner's g-prior: $\boldsymbol{\beta} \mid \sigma^2 \sim N(\mathbf{0}, g\sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1})$. When $g = n$, the covariance matrix is the same as the unit informative prior. Under the g-prior, we have

$$\begin{aligned} \mathrm{E}(\boldsymbol{\beta} \mid \boldsymbol{y}, \boldsymbol{X}, \sigma^2) &= \frac{g}{g+1}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y} \\ \mathrm{Var}(\boldsymbol{\beta} \mid \boldsymbol{y}, \boldsymbol{X}, \sigma^2) &= \frac{g}{g+1}\sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1} \end{aligned}$$

With an inverse-gamma for σ^2, the g-prior is a conjugate normal-inverse-gamma prior. Therefore, the posterior for $(\boldsymbol{\beta}, \sigma^2)$ can be expressed conditionally

$$\sigma^2 \mid \boldsymbol{y}, \boldsymbol{X} \sim \mathrm{IG}\left(\frac{\nu_0 + n}{2}, \frac{\nu_0\sigma_0^2 + \mathrm{SSR}_g}{2}\right)$$

where

$$\mathrm{SSR}_g = \boldsymbol{y}'\left[\boldsymbol{I} - \frac{g}{g+1}\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}\right]\boldsymbol{y}$$

which converges to SSR-OLS as $g \to \infty$.

$$\boldsymbol{\beta} \mid \boldsymbol{y}, \boldsymbol{X}, \sigma^2 \sim N\left(\frac{g}{g+1}\hat{\boldsymbol{\beta}}_{OLS}, \frac{g}{g+1}\sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}\right)$$

Posterior samples can be drawn sequentially.

Bayesian ridge regression. Consider the prior

$$\boldsymbol{\beta}|\sigma^2 \sim N(0, \sigma^2\tau^2\boldsymbol{I}),$$

which leads to posterior given σ^2: $\boldsymbol{\beta} \mid \boldsymbol{y}, \boldsymbol{X}, \sigma^2 \sim \text{MVN}(\boldsymbol{\beta}_n, \boldsymbol{\Sigma}_n)$, where

$$\begin{aligned} \boldsymbol{\beta}_n &= (\tau^{-2}\boldsymbol{I} + \boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y} \\ \boldsymbol{\Sigma}_n &= \sigma^2\left(\tau^{-2}\boldsymbol{I} + \boldsymbol{X}'\boldsymbol{X}\right)^{-1} \end{aligned}$$

The posterior mean $\boldsymbol{\beta}_n$ is exactly the ridge solution for linear regression. A full Bayesian model will incorporate a prior distribution, e.g., a half-Cauchy distribution, on the shrinkage parameter τ. This yields a shrinkage similar to ridge but with uncertainty over the penalty.

6.4 Classic Bayesian model selection

Now assume an indicator vector $\boldsymbol{z} = (z_1, \ldots, z_k)'$, where $z_j = 1$ means that the jth covariate is included in the model, $z_j = 0$ otherwise. The indicator vector $\boldsymbol{z}$ is latent (unknown) for model selection.

The Bayesian framework considers a joint prior for $(\boldsymbol{z}, \boldsymbol{\beta}, \sigma^2)$. Upon integrating out $(\boldsymbol{\beta}, \sigma^2)$, we can obtain the marginal likelihood for $\boldsymbol{z}$:

$$p(\boldsymbol{z} \mid \boldsymbol{y}, \boldsymbol{X}) = \frac{p(\boldsymbol{z})p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{z})}{\sum_{\boldsymbol{z}} p(\boldsymbol{z})p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{z})}$$

When we compare two models $\boldsymbol{z}_a$ vs. $\boldsymbol{z}_b$, the posterior odds ratio is

$$\frac{p(\boldsymbol{z}_a \mid \boldsymbol{y}, \boldsymbol{X})}{p(\boldsymbol{z}_b \mid \boldsymbol{y}, \boldsymbol{X})} = \frac{p(\boldsymbol{z}_a)}{p(\boldsymbol{z}_b)} \times \frac{p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{z}_a)}{p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{z}_b)}$$

The key to this model selection or comparison is the Bayes factor, which needs the marginal likelihood $p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{z})$

$$p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{z}) = \int_{\sigma^2} \int_{\boldsymbol{\beta}} p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{\beta}, \sigma^2, \boldsymbol{z}) p(\boldsymbol{\beta} \mid \boldsymbol{X}, \sigma^2, \boldsymbol{z}) p(\sigma^2 \mid \boldsymbol{z}) d\boldsymbol{\beta} d\sigma^2$$

However, computing this in a general model can be challenging.

We can obtain the closed-form expression under Zellner's g-prior for the regression model:

$$p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{z}) = \pi^{-\frac{n}{2}} \frac{\Gamma((\nu_0 + n)/2)}{\Gamma(\nu_0/2)} (1+g)^{-\frac{p_z}{2}} \frac{(\nu_0 \sigma_0^2)^{\nu_0/2}}{(\nu_0 \sigma_0^2 + \mathrm{SSR}_g^z)^{(\nu_0+n)/2}}$$

where p_z is the number of β's under model $\boldsymbol{z}$ and

$$\mathrm{SSR}_g^z = \boldsymbol{y}' \left[\boldsymbol{I} - \frac{g}{g+1} \boldsymbol{X}_z (\boldsymbol{X}_z' \boldsymbol{X}_z)^{-1} \boldsymbol{X}_z' \right] \boldsymbol{y}$$

where $\boldsymbol{X}_z$ is the reduced design matrix under model $\boldsymbol{z}$.

Other method: Bayesian information criteria (BIC). Based on the Taylor expansion, we have

$$p(y|\boldsymbol{X}, z) = \exp\left(\log \hat{L} - \frac{k_z}{2}\log(n) + O(1)\right)$$

Since

$$p(z|y, \boldsymbol{X}) \propto p(y|\boldsymbol{X}, z)p(z),$$

we can pick the model with the largest $\log \hat{L} - \frac{k_z}{2}\log(n) = -\text{BIC}/2$, or equivalently, the smallest BIC.

6.5 High-dimensional linear regression models

Recall that the OLS estimator minimizes the SSR($\boldsymbol{\beta}$):

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \text{SSR}(\boldsymbol{\beta}) = \arg\min_{\boldsymbol{\beta}} (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})$$

The lasso estimate is defined as

$$\hat{\boldsymbol{\beta}}^{lasso} = \arg\min_{\boldsymbol{\beta}} \left\{ \text{SSR}(\boldsymbol{\beta}) + \lambda \sum_{j=1}^{p} |\beta_j| \right\}$$

The only difference between the lasso and ridge estimators is that the the former uses ℓ_1 penalty, while the latter uses a ℓ_2 penalty. We prefer the lasso estimator in high-dimensional settings as it results in sparse solutions.

Tibshirani (1996) suggested that the lasso estimator can be interpreted as posterior mode estimates when the regression parameters have iid Laplace (i.e., double-exponential) priors. The posterior mode estimator is also called MAP (Maximum a posteriori estimation). Specifically,

$$\hat{\boldsymbol{\beta}}^{lasso} = \arg\max_{\boldsymbol{\beta}} p(\boldsymbol{\beta} \mid y, \boldsymbol{X}, \sigma^2, \tau),$$

when the prior distribution of $\boldsymbol{\beta}$ is

$$p(\boldsymbol{\beta}|\tau) = (\tau/2)^p \exp(-\tau\|\boldsymbol{\beta}\|_1).$$

For any fixed values σ^2 and τ, the posterior mode is the lasso estimate with $\lambda = 2\tau\sigma^2$.

The **Bayesian lasso** was motivated by a conditional Laplace prior, where

$$p(\beta_j|\sigma^2, \lambda) = \frac{\lambda}{2\sigma} \exp(-\lambda|\beta_j|/\sigma)$$

Conditioning on σ^2 is important as it ensures that the full posterior is unimodal. The above prior can be rewritten into the following hierarchical priors:

$$\beta_j \mid (\sigma^2, \gamma_j^2) \sim N(0, \sigma^2\gamma_j^2),\ \gamma_j^2 \sim Exp(\lambda^2/2).$$

Note that different texts use different parameterizations; here τ in the MAP-lasso discussed above is not the same as λ in the Bayesian-lasso hierarchical form.

Horseshoe / global-local shrinkage. This is typical for $p \gg n$, as it shrinks near 0 while keeping large signals. Specifically,

$$\beta_j \mid (\lambda_j, \tau, \sigma^2) \sim N(0, \sigma^2\tau^2\lambda_j^2),\ \lambda_j \sim \text{half-Cauchy},\ \tau \sim \text{half-Cauchy}$$

Spike-and-slab prior. It directly introduce a indicator for non-zero β_j:

$$\gamma_j \sim \text{Bernoulli}(p_0),\ \beta_j \mid \gamma_j \sim (1-\gamma_j)\delta_0 + \gamma_j N(0, \tau^2).$$

Posterior distributions for both models can be fitted through Gibbs sampling + slice sample/MH/RJ. For fast computation, mean-field variational Bayes (MFVB) is typically used to approximate the posterior distribution (will discuss the algorithm in a later chapter).

Implementation: the 'blasso' function in the R package *monomvn*.

Bayesian inference and selection: in high-dimensional Bayesian models, "selection" is not usually a single fixed set across MCMC draws. This is actually an advantage of Bayesian models, which reflects the model uncertainty. For example, if two correlated predictors can explain the same signal, the chain

may alternate: draw 1 includes X_1, excludes X_2; draw 2 excludes X_1, includes X_2.

For continuous shrinkage priors (Bayesian lasso, horseshoe), usually $\beta_j^{(m)} \neq 0$ almost surely for all draws, so "selection" is not automatic and an extra selection rule needs to be specified. For example, the posterior probability of "practical relevance" is commonly defined based on a pre-specified threshold value δ.

For the spike-and-slab prior, each draw includes latent indicators $\gamma_j^{(m)} \in \{0, 1\}$, so the coefficients can be exactly zero. Posterior inclusion probability for variable j ($\hat{\pi}_j$) is usually reported.

Variables are selected according to $\hat{\pi}_j$ (e.g., threshold $\hat{\pi}_j$ or control Bayesian FDR). Posterior median or conditional posterior mean are usually reported for the selected variables, along with joint uncertainties (e.g., model size distribution, top models, and co-inclusion matrix).

7 Hamiltonian Monte Carlo

7.1 Hamiltonian dynamics

Hamiltonian dynamics has a physical interpretation that can provide useful intuitions. Suppose that there is a frictionless ball sliding over a surface of varying height. The state of this system consists of the *position* of the ball, given by a d-dimensional vector q, and the *momentum* of the ball (mass times velocity), given by a d-dimensional vector p. The *potential energy*, $U(q)$, of the ball is proportional to the height of the surface and its current position, and its *kinetic energy*, is equal to $||p||^2/(2m)$. On a level part of the surface, the ball moves at a constant velocity, equal to $||p||/m$; at a rising slope, the ball's momentum allows it to continue, with its kinetic energy decreasing and its potential energy increasing, until the kinetic energy (and hence p) is zero, at which point it will slide back down.

Formally, the *Hamiltonian*, denoted as $H(q,p) = U(q) + K(p)$, is the system that operates on a d-dimensional position vector q and a d-dimensional momentum vector p. The partial derivatives over time t of the Hamiltonian determine how q and p change over time. According to Hamilton's equations, we have

$$\frac{dq_i}{dt} = \frac{\partial H}{\partial p_i}, \ \frac{dp_i}{dt} = -\frac{\partial H}{\partial q_i}, \tag{1}$$

for $i = 1, \cdots d$.

The properties of Hamilton dynamics are:

1. Reversibility: the mapping T_s from the state at time t, $(q(t), p(t))$, to the state at $t+s$, $(q(t+s), p(t+s))$, is one-to-one, and hence has an inverse, T_{-s}.

2. Invariant:

$$\frac{dH}{dt} = \sum_{i=1}^{d} \left[\frac{dq_i}{dt}\frac{\partial H}{\partial q_i} + \frac{dp_i}{dt}\frac{\partial H}{\partial p_i} \right] = \sum_{i=1}^{d} \left[\frac{\partial H}{\partial p_i}\frac{\partial H}{\partial q_i} - \frac{\partial H}{\partial q_i}\frac{\partial H}{\partial p_i} \right] = 0.$$

   This is important because part of HMC involves a Metropolis step with proposal found by Hamiltonian dynamics. The invariant property makes the acceptance probability close to 1.

3. Volume preservation: if we apply the mapping T_s to the points in some region R of (q, p) space, with volume V, the image of R under T_s will also have volume V. This is important because we do not need to account for any change in volume in the acceptance probability for Metropolis updates.

7.2 Discretizing Hamilton's equations: the Leapfrog method

For computer implementation, Hamilton's equations must be approximated by discretizing time, using some small stepsize, ϵ. Starting with the state at time zero, we iteratively compute (approximately) the state at times $\epsilon, 2\epsilon, 3\epsilon, \ldots$

The most commonly used method to approximate the solution to a system of

differential equations is Euler's method:

$$\begin{aligned} p_i(t+\epsilon) &= p_i(t) + \epsilon \frac{dp_i}{dt}(t) = p_i(t) - \epsilon \frac{\partial U}{\partial q_i}(q(t)) \\ q_i(t+\epsilon) &= q_i(t) + \epsilon \frac{dq_i}{dt}(t) = q_i(t) + \epsilon \frac{\partial U}{\partial p_i}(p(t)) \end{aligned}$$

However, the method does not work well.

The Leapfrog method is a modification of Euler's method and proved better performance:

$$\begin{aligned} p_i(t+\epsilon/2) &= p_i(t) - (\epsilon/2) \frac{\partial U}{\partial q_i}(q(t)) \\ q_i(t+\epsilon) &= q_i(t) + \epsilon \frac{\partial U}{\partial p_i}(p(t+\epsilon/2)) \\ p_i(t+\epsilon/2) &= p_i(t+\epsilon/2) - (\epsilon/2) \frac{\partial U}{\partial q_i}(q(t+\epsilon)) \end{aligned}$$

That is, We start with a half step for the momentum variables, then do a full step for the position variables, using the new values of the momentum variables, and finally do another half step for the momentum variables, using the new values for the position variables.

The Leapfrog method preserves the reversibility and volume preservation properties stated in the previous section. The Leapfrog preserves the invariant property when $\epsilon \to 0$. That is, when we apply the Leapfrog to HMC, there exists a trade-off between acceptance rate and computation complexity (number of log-likelihood and gradient evaluations).

7.3 HMC

The distribution we wish to sample can be related to a potential energy function via the concept of a canonical distribution from statistical mechanics. Given some energy function, $E(x)$, for the state x, of the some physic system, the canonical distribution over states has pdf/pmf

$$P(x) = \frac{1}{Z} \exp\{-E(x)/T\},$$

where T is the temperature of the system (not very important for HMC) and Z is the normalizing constant. Thus, to obtain $P(x)$, with $T = 1$, we should set the energy function to be

$$E(x) = -\log P(x) - \log Z \tag{2}$$

The Hamiltonian is an energy function for the joint state position q and momentum p. In MCMC applications of Hamiltonian dynamics, i.e., Hamiltonian Monte Carlo (HMC), the position q will correspond to the variable of interest θ. The momentum variable p, one for each dimension of θ, will be introduced artificially to allow Hamiltonian dynamics to operate. Motivated by (2), We usually assume

$$H(q, p) = U(q) + K(p),$$

which results in the joint density

$$P(q, p) = \frac{1}{Z} \exp\{-U(q)/T\} \exp\{-K(p)/T\} \tag{3}$$

Here, the potential energy $U(q)$ will be defined as

$$U(q) = -\log[p(\theta|y)] = -\log[\pi(q)L(q|D)],$$

and the kinetic energy is usually defined as

$$K(p) = p^T M^{-1} p,$$

which corresponds to the negative log-likelihood of the zero-mean Gaussian distribution with covariance matrix M.

HMC samples from the canonical distribution for q and p defined in (3), which involves in two steps:

1. new values for the momentum variables are randomly drawn from their Gaussian distribution $N(0, M)$, independently of the current values of the position variables

2. Metropolis update using Hamiltonian dynamics: starting current state, (q, p), Hamiltonian dynamics is simulated for L steps using the Leapfrog method with a stepsize of ϵ. The momentum variables at the end of this L-step trajectory are then negative, giving a proposed state (q^*, p^*). The proposed state is accepted as the next state of the Markov chain with acceptance probability

$$\min[1, \exp\{-U(q^*) + U(q) - K(p^*) + K(p)\}]$$

Good properties of HMC:

1. Ergodicity: will not be trapped in some subset of state space
2. No random work: uses gradient of log-likelihood (energy) for more efficient sampling

Demos: https://chi-feng.github.io/mcmc-demo/

7.4 Stan

Stan is the most popular probabilistic programming framework. The key algorithm for Stan is the NUTS (No-U-Turn Sampler), a dynamic HMC sampler. The basic HMC algorithm introduced in the previous section is static HMC, where the number of steps L is fixed. NUTS is a dynamic HMC algorithm, where dynamic refers to the dynamic trajectory length. The dynamic is achieved through including the slice sampling into the Leapfrog discretization. We will ignore the details of NUTS.

Examples of Stan codes are available on Canvas. You can write in Stan language any program to compute the log density. Stan compiles the model written in Stan language to C++. To operate Stan in R,

1. *RStan* is an R interface that directly interfacing the C++ that Stan is built on top of with R
2. The R package *CmdStanR* is an alternative R interface for Stan (usually recommended by Stan developers)
3. *RStanARM* provides simplified model description with pre-compiled models (restricted)
4. *brms* provides a simplified model description with a larger set of models than *RStanARM*; need to wait for the compilation (much slower than RStanARM)

Some useful resources for learning Stan:

1. Introductory article in Journal of Statistical Software
2. Stan home page
3. RStan installation
4. CmdStanR installation
5. YouTube video: "Basics of Bayesian inference and Stan

CmdStanR Example: Bernoulli model

Code

```
data {
  int<lower=0> N;
  array[N] int<lower=0, upper=1> y;
}
parameters {
  real<lower=0, upper=1> theta;
}
model {
  theta ~ beta(1, 1); // uniform prior on interval 0,1
  y ~ bernoulli(theta);
}
```

1. Data type and size need to be declared
2. Only continuous parameters allowed
3. model: "∼" defines a distribution statement; it can also be written as log density increment statements ("_lpdf" for continuous, "_lpmf" for discrete):

Code

```
model {
  target += beta_lpdf(theta | 1, 1);
  target += bernoulli_lpmf(y | theta);
}
```

Compile Stan model:

Code

```
data_list<-list(N = 10, y = c(0,1,0,0,0,0,0,0,0,1))
fit<-mod$sample(
  data = data_list,
  seed = 123,
  chains = 4,
  parallel_chains = 4,
  refresh = 500 # print update every 500 iters
)

fit$summary()

# HMC specified diagnosis
fit$diagnostic_summary(diagnostics = c("divergences",
                                       "treedepth"))

mcmc_hist(fit$draws("theta"))
```

CmdStanR Example

Code

```
library(rstanarm)
dat<-data.frame(N = c(674,680), y = c(39,22), grp = c(0,1))
fit2<-stan_glm(y/N~grp, weights = N, family = binomial(), data=dat)
```

brms Example

Code

```
library(brms)
dat<-data.frame(N = c(674,680), y = c(39,22), grp = c(0,1))
fit3<-brm(y|trials(N)~grp, family = binomial(), data = dat)
```

8 Hierarchical modeling

8.1 Comparing multiple groups

Example: the data "nels.RData" is part of the 2002 Educational Longitudinal Study (ELS), a survey of students from a large sample of schools across the United States. This dataset includes a population of schools as well as a population of students within each school. Datasets like this, where there is a hierarchy of nested populations, are often called *hierarchical* or *multilevel*.

Review of ANOVA

Marginal vs. conditional vs. individual models

Exchangeability and hierarchical models. Recall that exchangeability $P(y_1, \ldots, y_n) = P(y_{\pi_1}, \ldots, y_{\pi_n})$ is equivalent to

$$\{y_1, \ldots, y_n \mid \phi\} \overset{i.i.d.}{\sim} p(y \mid \phi), \qquad \phi \sim p(\phi)$$

Let us now consider data $\{\boldsymbol{y}_1, \ldots, \boldsymbol{y}_m\}$ from m groups, where in each group, $\boldsymbol{y}_j = \{y_{1j}, \ldots, y_{n_j,j}\}$. The total sample size is $N = \sum_j n_j$.

Under exchangeability, we can model data in the jth group as

$$\{y_{1j}, \ldots, y_{n_j,j} \mid \phi_j\} \overset{i.i.d.}{\sim} p(y \mid \phi_j)$$

where ϕ_j is called the group-specific parameter.

We do not treat $\phi_1, \ldots, \phi_m$ as completely independent, but we believe that they are exchangeable. Therefore

$$\{\phi_1, \ldots, \phi_m \mid \psi\} \overset{i.i.d.}{\sim} p(\phi \mid \psi)$$

where ψ is called a hyper-parameter.

Now we can build a hierarchical model (also called a multi-level model):

$$\begin{aligned} y_{ij} &\overset{i.i.d.}{\sim} p(y \mid \phi_j), \quad i = 1, \ldots, n_j \\ \phi_j &\overset{i.i.d.}{\sim} p(\phi \mid \psi), \quad j = 1, \ldots, m \\ \psi &\sim p(\psi) \end{aligned}$$

Specifically, in the ANOVA setting, the hierarchical normal model is specified as follows

$$\begin{aligned} y_{ij} \mid \theta_j, \sigma^2 &\sim N(\theta_j, \sigma^2) \\ \theta_j \mid \mu, \tau^2 &\sim N(\mu, \tau^2) \\ \sigma^2 &\sim IG(\nu_0/2, \nu_0\sigma_0^2/2) \\ \tau^2 &\sim IG(\eta_0/2, \eta_0\tau_0^2/2) \\ \mu &\sim N(\mu_0, \gamma_0^2) \end{aligned}$$

Other hierarchical model: Binomial model

$$\begin{aligned} y_{ij} \mid \theta_j, n_j &\sim \text{Bin}(n_j, \theta_j) \\ \theta_j \mid \alpha, \beta &\sim \text{Beta}(\alpha, \beta) \\ p(\alpha, \beta) &\propto (\alpha + \beta)^{-5/2} \end{aligned}$$

The joint hyperprior is called the diffuse prior for location and scale parameters (BDA3 p.110)

8.2 Posterior inference: Gibbs sampler

In this normal hierarchical model, the unknown parameters include $\{\phi_1, \ldots, \phi_m\}, \sigma^2, \mu$ and τ^2. The posterior distribution can be obtained by

$$\begin{aligned} p(\{\phi_1, \ldots, \phi_m\}, \sigma^2, \mu, \tau^2 \mid y) &\propto p(y \mid \boldsymbol{\theta}, \sigma^2) p(\boldsymbol{\theta} \mid \mu, \tau^2) p(\sigma^2) p(\mu) p(\tau^2) \\ &= \left\{ \prod_{j=1}^{m} \prod_{i=1}^{n_j} p(y_{ij} \mid \theta_j, \sigma^2) \right\} \left\{ \prod_{j=1}^{m} p(\theta_j \mid \mu, \tau^2) \right\} p(\sigma^2) p(\mu) p(\tau^2) \end{aligned}$$

To sample from this high-dimensional distribution, we use Gibbs sampling. Derive the full conditional distributions as follows.

1. $p(\mu \mid \text{others}) \propto p(\mu) \prod_{j=1}^{m} p(\theta_j \mid \mu, \tau^2)$ which is

$$N\left(\frac{m\bar{\theta}/\tau^2 + \mu_0/\gamma_0}{m/\tau^2 + 1/\gamma_0^2}, \frac{1}{m/\tau^2 + 1/\gamma_0^2} \right)$$

2. $p(\tau^2 \mid \text{others}) \propto p(\tau^2) \prod_{j=1}^{m} p(\theta_j \mid \mu, \tau^2)$ which is

$$\text{IG}\left(\frac{\eta_0 + m}{2}, \frac{\eta_0 \tau_0^2 + \sum_{j=1}^{m} (\theta_j - \mu)^2}{2}\right)$$

3. $p(\theta_j \mid \text{others}) \propto p(\theta_j \mid \mu, \tau^2) \prod_{i=1}^{n_j} p(y_{ij} \mid \theta_j, \sigma^2)$ which is

$$N\left(\frac{n_j \bar{y}/\sigma^2 + \mu/\tau^2}{n_j/\sigma^2 + 1/\tau^2}, \frac{1}{n_j/\sigma^2 + 1/\tau^2}\right)$$

4. $p(\sigma^2 \mid \text{others}) \propto p(\sigma^2) \left\{\prod_{j=1}^{m} \prod_{i=1}^{n_j} p(y_{ij} \mid \theta_j, \sigma^2)\right\}$ which is

$$\text{IG}\left(\frac{\nu_0 + \sum_{j=1}^{m} n_j}{2}, \frac{\nu_0 \sigma_0^2 + \sum_j \sum_i (y_{ij} - \theta_j)^2}{2}\right)$$

Under the classical ANOVA framework, even though θ_j can be a random effect as in the Bayesian model, the variance parameter σ^2 has to be assumed common across groups. Under the Bayesian framework, however, we can have more flexibility in modeling.

Let σ_j^2 be the variance of group j, which is heterogeneous across groups. We can modify the hierarchical normal model as

$$y_{ij} \mid \theta_j, \sigma_j^2 \sim N(\theta_j, \sigma_j^2), \quad \sigma_j^2 \sim \text{IG}(\nu_0/2, \nu_0\sigma_0^2/2)$$

Then, two steps in the Gibbs sampling will be changed

$$\theta_j \mid \text{others} \sim N\left(\frac{n_j\bar{y}/\sigma_j^2 + \mu/\tau^2}{n_j/\sigma_j^2 + 1/\tau^2}, \frac{1}{n_j/\sigma_j^2 + 1/\tau^2}\right)$$

$$\sigma_j \mid \text{others} \sim \text{IG}\left(\frac{\nu_0 + n_j}{2}, \frac{\nu_0\sigma_0^2 + \sum_i (y_{ij} - \theta_j)^2}{2}\right)$$

If ν_0 and σ_0^2 are assumed unknown, priors can be specified on them as well. The textbook suggests

$$p(\sigma_0^2) \sim \text{gamma}(a, b)$$
$$p(\nu_0) \propto e^{-\alpha\nu_0}$$

a discrete geometric distribution on $\{1, 2, \ldots\}$. The full conditional distributions of these two parameters are given by

$$p(\sigma_0^2 \mid \text{others}) \sim \text{gamma}(a + m\nu_0/2, b + \sum_{j=1}^{m} (\sigma_j^2)^{-1}/2)$$

$$p(\nu_0 \mid \text{others}) \propto \left(\frac{\left(\frac{\nu_0 \sigma_0^2}{2}\right)^{\frac{\nu_0}{2}}}{\Gamma(\nu_0/2)} \right)^m \left(\prod_{j=1}^m \frac{1}{\sigma_j^2} \right)^{\nu_0/2-1} \exp\left\{ -\nu_0 \left(\alpha + \frac{\sigma_0^2}{2} \sum_{j=1}^m \frac{1}{\sigma_j^2} \right) \right\}$$

The above density is not commonly recognized, and henc, an MH algorithm is needed within the Gibbs cycle.

8.3 Posterior inference: Stan

When HMC (or Stan) is considered, conjugacy no longer matters and any convenient prior distributions can be used. For example, by restricting the parameter space to be strictly positive, we can simply assume

$$\sigma^2 \sim N(0,1) \text{ and } \tau^2 \sim N(0,1),$$

which is equivalent to a half-normal distribution. The corresponding Stan codes are:

Code

```
data {
  int<lower=1> N_obs;
  int<lower=1> N_school;
  vector[N_obs] score;
  array[N_obs] int<lower=1, upper=N_school> school;
}
parameters {
  vector[N_school] mu_school;
  real mu_0;
```

```
  real<lower=0> sd;
  real<lower=0> sigma;
}
model {
  for (j in 1:N_school){
    mu_school[j] ~ normal(mu_0, sd);
  }
  mu_0 ~ normal(0, 1);
  sd ~ normal(0, 1);
  sigma ~ normal(0, 1);

  for(i in 1:N_obs){
    score[i] ~ normal(mu_school[school[i]], sigma);
  }
}
```

We can also use the *brms* package:

Code

```
fit2<-brm(mathscore ~ 1 + (1 | school), data=dat,
        prior = c(prior(normal(0,1), class="Intercept"),
                  prior(normal(0,1), class="sd"),
                  prior(normal(0,1), class="sigma")))
```

Other commonly used models:

Code

```
#Heteroskedastic variance:
bf(y ~ x, sigma ~ x)
#t-distribution:
y ~ x, family = student()
#Logistic regression:
y|trials(N) ~ x, family = binomial()
```

```
#Negative binomial:
y ~ x; family = negbinomial()
```

9 Asymptotic and frequentist properties

9.1 Normal approximation

If $p(\theta \mid y)$ is unimodal and roughly symmetric, it is convenient to approximate it by a normal distribution (Laplace approximation). More elaborate approximation algorithms will be discussed in the next chapter.

Review: Laplace approximation
The key idea of the Laplace approximation is to approximate the log-posterior density by a quadratic function. Let $h(\theta)$ be the log-posterior density function:

$$h(\theta) := h(\theta|y) = \log(p(y \mid \theta)) + \log(p(\theta)) + C$$

Apply a second-order Taylor expansion at its mode: $\hat{\theta}$

$$h(\theta) \approx h(\hat{\theta}) + (\theta - \hat{\theta})' h''(\hat{\theta})(\theta - \hat{\theta})/2$$

where h'' is the Hessian matrix of h. Since this is the logarithm of the target posterior, let's now transform it exponentially back:

$$e^{h(\hat{\theta})} e^{(\theta-\hat{\theta})' h''(\hat{\theta})(\theta-\hat{\theta})/2}$$

which implies a Gaussian (normal) density with mean vector $\hat{\theta}$ and covariance matrix $[-h''(\hat{\theta})]^{-1}$.

$I_n(\theta) = -h''(\theta)$ is the observed information matrix, which measures curvature of the posterior density. $I_n(\hat{\theta})$ is positive definite when $\hat{\theta}$ is in the interior of the parameter space.

If θ is high-dimensional, it is possible that θ can be partitioned into two subvectors, i.e., $\theta = (\theta_1, \theta_2)$, and 1. may be only $\theta_1|(\theta_2, y)$ can be well-approximated by a normal distribution; 2. θ_1 is a vector of low-dimensional targeted parameters, and θ_2 are high-dimensional nuisance parameters.

Example: causal inference; semiparametric models

9.2 Asymptotic normality and consistency

Under regularity conditions, e.g., continuity and moment conditions, the posterior distribution of θ approaches normality with mean θ_0 and variance $[nI(\theta_0)]^{-1}$ as $n \to \infty$ (not a formal asymptotic statement), where

$$I(\theta_0) = -\mathrm{E}\left\{\frac{\partial^2}{\partial\theta^2}\log p(Y|\theta) \mid \theta = \theta_0\right\}$$

is the Fisher information matrix. As $[nI(\theta_0)]^{-1} \to 0$, $\hat{\theta} \xrightarrow{p} \theta$.

$n^{-1}I_n(\hat{\theta})$ consistently estimate $I(\theta_0)$.

The asymptotic results indicate that the importance of the prior distribution diminishes as the sample size increases. However, when sample size is small, the prior distribution is a critical part of the model specification.

The R package *priorsense* can be used to check the prior and likelihood sensitivity. The package works with models created with rstan, cmdstanr, brms, R2jags, or with draws objects from the posterior package (https://github.com/n-kall/priorsense).

The sensitivity check modifies the posterior by power-scaling the prior or likelihood slightly:

$$p(\theta)^{\alpha} p(y|\theta) \text{ or } p(\theta) p(y|\theta)^{\alpha}$$

The package is computationally efficient as it uses Pareto-smoothed importance sampling to estimate modified posterior from the original posterior draws.

9.3 Counterexamples

Under certain situations, the large sample theorem, or even the model itself, may fail:

1. Under-identified models or non-identifiable parameters (wrong model or poor data set)
2. High-dimensional data
3. Unbounded likelihood or heavy-tailed density function
4. Improper posterior distributions (usually caused by an improper prior distribution)
5. Prior distributions that exclude the true value

6. Convergence to the edge of parameter space

9.4 Bayesian hypothesis testing

Most common statistical tests can be converted to linear models and fitted using Stan.

RStanARM examples:

Code

```
#One-sample t-test
stan_glm(y ~ 1)

#Paired t-test
stan_glm((y1 - y2) ~ 1)

#Two-sample t-test
stan_glm(y ~ 1 + group)

#Pearson correlation
stan_glm(y ~ 1 + x)
```

The Bayesian posterior mean (median) and credibility interval analogs the corresponding Frequentist point estimates and confidence interval.

The frequentist hypothesis testing asks what happens if data is generated from the null model, but could not tell whether the alternative model is good enough.

For Bayesian hypothesis testing, people usually report the full posterior and compare the distribution to "expert information".

For continuous posterior distributions, there is zero probability that, e.g., the treatment effect is exactly 0.

When we know the sign of the effect, we can directly calculate the corresponding posterior probability.

Some people also make the decision by comparing whether the credibility interval includes the null case.

Some people also define the "region of practical equivalence" (ROPE), usually of the form

$$P(\theta_0 - \epsilon < \theta < \theta_0 + \epsilon \mid y),$$

where ϵ is a pre-specified practically useful effect size.

Bayes factor, which is defined as

$$BF = \frac{p(\theta_1 \mid y)}{p(\theta_2 \mid y)},$$

can also be used for Bayesian hypothesis testing. A point mass at the null value should be introduced on the prior distribution (sensitive to the prior choice)

10 Approximation algorithms

10.1 Optimization algorithms

We have talked about using the posterior mode as a point estimate for the parameter of interest, e.g., Laplace approximation. We first discuss mode optimization algorithms.

Newton's method, or the Newton-Raphson algorithm, is the most commonly used method for optimization. It is an iterative approach based on a second-order Taylor approximation of the log-posterior density $h(\theta)$

The method does not require $h(\theta)$ to be normal, but it requires certain local smoothness and boundedness conditions.

With the previous value θ^{t-1}, θ^t is obtained by

$$\theta^t = \theta^{t-1} - [h''(\theta^{t-1})]^{-1} H'(\theta^{t-1})$$

The method is also called the gradient descent or steepest descent, as the solution moves in the negative gradient direction to ensure the fastest decrease in function value

The method is sensitive to the choice of initial value θ^0. If it works, the algorithm converges to the maximum (can be local) very fast when the iterations are close to the solution (quadratic approximation is accurate).

Both the gradient ($h'(\theta)$) and Hessian ($h''(\theta)$) can be approximated numerically.

Quasi-Newton methods. Since Hessian is a $p \times p$ matrix, approximating it can be computationally intensive (also costs a huge amount of memory for storage). Quasi-Newton methods, such as the Broyden-Fletcher-Goldfarb-Shanno (BFGS) method, form an approximation of the Hessian using only gradient information (less accurate).

Conjugate gradient (CG) methods. Instead of moving in the negative gradient direction, CG chooses search directions that are orthogonal under the Hessian inner product. CG is likely to require more iterations than Newton and quasi-Newton methods, but uses less computation per iteration and requires less storage.

Finding multiple modes. A simple way to search for multiple modes is to start the iterations at many points throughout the parameter space.

10.2 Laplace's methods

We have discussed the Laplace approximation in previous chapters, where the posterior distribution is approximated with normal distribution. We can also use Laplace's method to approximate expected values, $\mathrm{E}(g(\theta) \mid y)$, where $g()$ is a smooth function.

Suppose K modes are found (with different initial values). The posterior distribution can then be approximated by a mixture of K multivariate normal distributions, each with its own mode $\hat{\theta}_k$, variance V_k, relative mass w_k.

For each k, if the modes are fairly widely separated and the normal approximation is appropriate for each mode, the mass w_k can be estimated by equating the unnormalized posterior density to the approximation.

One can also replace each normal density by a multivariate t with some small number of degrees of freedom ν.

10.3 EM algorithm

In problems with many parameters, normal approximations to the joint distribution are often useless, and the joint mode is typically not helpful. Let $\theta = (\phi, \gamma)$, it is often useful, however, to base an approximation on a marginal posterior mode of a subset of the parameter, i.e., $p(\phi|y)$. After approximating $p(\phi|y)$ as a normal (or t, or a mixture), we can subsequently approximate $p(\gamma \mid \phi, y)$ as normal (or t, or a mixture).

The EM algorithm is an iterative method for finding the mode of the marginal posterior density $p(\phi|y)$, when maximizing $p(\phi|y)$ directly is hard but easy to work with $p(\gamma \mid \phi, y)$ and $p(\phi \mid \gamma, y)$.

The EM algorithm was originally proposed for MLE with missing values. In our context, γ can be considered as missing data. One example is the hierarchical model, where $\gamma = (\theta_1, \cdots, \theta_m)$ and $\phi = (\mu, \tau^2, \sigma^2)$.

The EM algorithm is based on a simple identity:

$$p(\gamma, \phi \mid y) = p(\gamma \mid \phi, y)p(\phi \mid y) \implies \log p(\phi \mid y) = \log p(\gamma, \phi \mid y) - \log p(\gamma \mid \phi, y)$$

E step: expectation. Given the current state t (ϕ^t), take expectations with respect to "missing values" γ:

$$\log p(\phi^t \mid y) = \mathrm{E}_\gamma\{\log p(\gamma, \phi^t \mid y)\} - \mathrm{E}_\gamma\{\log p(\gamma \mid \phi^t, y)\}$$

The second term is nuisance, as it is maximized at ϕ^t.

M step: maximization. Maximize the first term, usually called $Q(\phi|\phi^t)$, in the above equation:

$$\phi^{t+1} = \arg\max_\phi Q(\phi|\phi^t) = \arg\max_\phi \int [\log p(\gamma, \phi \mid y)]p(\gamma \mid \phi^t, y)d\gamma$$

GEM (generalized EM) algorithm. Instead of maximizing $Q(\phi|\phi^t)$, only find ϕ^{t+1} that increases the function.

The algorithm guaranties $\log p(\phi^{t+1} \mid y) > \log p(\phi^t \mid y)$

Debugging. A useful debugging check when running an EM algorithm is to compute the logarithm of the marginal (unnormalized) posterior density $\log p(\phi^t \mid y)$ at each iteration, and check that it increases monotonically.

MCEM algorithm. When there is no analytic solution in the E-step, the MCEM algorithm generates samples from the conditional distribution, $p(\gamma \mid \phi^t, y)$, and approximates the expectation by averaging over these Monte Carlo samples. If direct sampling is also not available, then MCMC are commonly used.

MCEM is a special case of the stochastic EM (SEM) algorithm. The main idea is to replace the E step (sometimes also the M step) with a less efficient but more computationally feasible method. The stochastic gradient descent algorithm, the most commonly used algorithm in deep learning, follows a similar idea.

ECM algorithm. The ECM algorithm is a variant of the EM algorithm in which the E-step is unchanged but the M-step is replaced by S conditional maximizations, or CM-steps. That is, the next state ϕ^{t+1} is determined by a sequence $\{\phi^{t+s/S}\}_{s=1}^{S}$. At the sth CM step, $\phi^{t+s/S}$ is determined by maximizing the $Q(\phi|\phi^t)$, given a pre-specified constraint function $g_s()$. By partition ϕ into S subvectors, the most common choice of $g_s()$ is constrain all parameters except ϕ_s to equal their current values.

Alternative ECM (AECM) algorithm. The ECM algorithm can be further generalized by allowing different alternating definitions of γ at each conditional maximization step. This generalization is most straightforward when γ represents missing data, and where there are different ways of completing the data at different maximization steps.

Supplemental EM/ECM algorithm for asymptotic variance estimation. Let $M(\phi)$ be the mapping defined implicitly by the EM algorithm, i.e., $\phi^{t+1} = M(\phi^t)$, the asymptotic variance matrix V, defined at the marginal mode $\hat{\phi}$, is

$$V = V_c + V_c J_M (I - J_M)^{-1},$$

where J_M is the Jacobian matrix for the EM map $M(\phi)$ evaluated at $\hat{\phi}$, and

$$V_c = \left[\mathrm{E}_\gamma \left(-\frac{d^2}{d\theta^2} \log p(\phi, \gamma \mid y) \middle| \phi, y \right) \middle|_{\phi=\hat{\phi}} \right]^{-1},$$

is the asymptotic variance matrix based on the logarithm of the joint posterior average over γ.

V_c can be computed analytically. The matrix J_M is computed numerically using an EM-type algorithm based on ϕ^t, ϕ^{t+1}, and $\hat{\phi}$.

For the ECM algorithm, the asymptotic variance is

$$V = V_c + V_c(J_M^{\text{EC}} - J_M^{\text{C}})(I - J_M^{\text{EC}})^{-1},$$

where J_M^{EC} defined in a manner analogous to J_M with ECM in place of EM, and J_M^{C} is the rate of convergence for conditional maximization directly to $\log p(\phi|y)$. Let $\nabla_s = \nabla g_s(\hat{\phi})$, the latter matrix is

$$J_M^{\text{C}} = \prod_{s=1}^{S} \left\{ \nabla_s (\nabla_s^\top V_c \nabla_s)^{-1} \nabla_s^\top V_c \right\}$$

Example: find $p(\mu, \tau, \sigma \mid y)$ in the hierarchical normal model. The marginal posterior is:

$$p(\mu, \log\tau, \log\sigma \mid y) \propto \tau \prod_{j=1}^{m} N(\hat{\theta}_j \mid \mu, \tau^2) \prod_{j=1}^{m} \prod_{i=1}^{n_j} N(y_{ij} \mid \hat{\theta}_j, \sigma^2) \prod_{j=1}^{m} V_{\theta_j},$$

where $V_{\theta_j} = (1/\tau^2 + n_j/\sigma^2)^{-1}$ and $\hat{\theta}_j = V_{\theta_j}(\mu/\tau^2 + n_j\bar{y}_j/\sigma^2)$ are the posterior variance and mean of θ_j, respectively. Due to the conjugacy, the result is exact.

To obtain the mode of the above marginal posterior using EM algorithm, we average over the parameters $\theta_1, \cdots, \theta_m$ in the E-step and maximize over $(\mu, \log\tau, \log\sigma)$ in the M-step. The log-posterior is:

$$\begin{aligned}\log p(\mu, \log\tau, \log\sigma \mid y) &= -n\log\sigma - (m-1)\log\tau - \frac{1}{\tau^2}\sum_{j=1}^{m}(\theta_j - \mu)^2 - \\ &\quad \frac{1}{2\sigma^2}\sum_{j=1}^{m}\sum_{i=1}^{n_j}(y_{ij} - \theta_j)^2 + \text{const.}\end{aligned}$$

E step: due to the normality of the conditional posterior, we can obtain the conditional expectation explicitly:

$$\mathrm{E}_\theta(\theta_j - \mu)^2 = (\hat{\theta}_j - \mu)^2 + V_{\theta_j}, \;\; \mathrm{E}_\theta(y_{ij} - \theta)^2 = (y_{ij} - \theta_j)^2 + V_{\theta_j}$$

M step.

$$
\begin{aligned}
\mu^{t+1} &= \frac{1}{m}\sum_{j=1}^{m}\hat{\theta}_j \\
\sigma^{t+1} &= \left\{\frac{1}{N}\sum_{j=1}^{m}\sum_{i=1}^{n_j}\left((y_{ij}-\hat{\theta}_j)^2 + V_{\theta_j}\right)\right\}^{1/2} \\
\tau^{t+1} &= \left\{\frac{1}{m-1}\sum_{j=1}^{m}\left((\hat{\theta}_j-\mu^{t+1})^2 + V_{\theta_j}\right)\right\}^{1/2}
\end{aligned}
$$

Note: $\hat{\theta}_j$ and V_{θ_j} should be updated for each iteration (can also be denoted as $\theta_j(\phi^t)$ and $V_{\theta_j}(\phi^t)$).

10.4 Variational inference

Variational Bayes (VB) is an algorithmic framework, similar to EM, for approximating a joint distribution. EM proceeds by alternately evaluating conditional expectations of the log density and using these to maximize a function of a set of parameters, converging to a point estimate (mode) of the parameters and thus an approximation to the posterior distribution.

In VB, the iterations lead to a closed-form approximation that is the closest fit to the posterior distribution within a pre-specified class of functions.

Similar to Laplace and EM, VB is typically applied in settings where we cannot easily take posterior draws from $p(\theta \mid y)$, nor can we easily compute expectations of interest

In VB, we approximate the posterior $p(\theta \mid y)$ by a simpler distribution $g(\theta)$. Given a pre-specified distribution class of g, we optimize g by maximizing the *evidence lower bound (ELBO)*:

$$\begin{aligned} \text{ELBO}(g) &= \text{E}_g[\log p(y,\theta)] - \text{E}_g[\log g(\theta)] \\ &= \log p(y) - \text{KL}(q||p). \end{aligned}$$

Thus, maximizing ELBO is equivalent to minimizing the Kullback-Leibler (KL) divergence:

$$\text{KL}(g||p) = \text{E}_g\left(\log\left(\frac{p(\theta \mid y)}{g(\theta)}\right)\right) = \int \log\left(\frac{p(\theta \mid y)}{g(\theta)}\right) g(\theta) d\theta$$

When $g(\theta|\phi)$ is Gaussian, where $\phi = (\mu, \tau^2)$ are denoted as hyperparameters of VB, the Laplace approximation is similar to VB (although in general, the Gaussian produced by Laplace is usually not the same as the Gaussian produced by Gaussian VB).

VB is a flexible framework so that any q function can be used as long as it does not alter the original parameter space. However, as it is commonly used for approximating the posterior of complex models, the most popular class

is the "Mean Field family", and the corresponding procedure is called *Mean Field Variational Inference (MFVI)*, which constrains the components of θ to be independent:

$$g(\theta \mid \phi) = \prod_{j=1}^{J} g_j(\theta_j \mid \phi_j),$$

where $\theta = (\theta_1, \cdots, \theta_J)$.

In MFVI, when working with exponential family models with conditionally conjugate priors, g_j can typically be determined by inspection, and the necessary expectations can be calculated in closed form. In nonconjugate models, MFVI can still be done by working within restricted functional forms such as normal distributions.

The coordinate Ascent Variational Inference (CAVI) is the algorithm typically

used for solving MFVI. For each θ_j, we have $g(\theta) = g_j(\theta_j)g_{-j}(\theta_{-j})$ and

$$\begin{aligned}
-\text{ELBO}(g) &= \int\int g_j(\theta_j)g_{-j}(\theta_{-j})\left(-\log p(y,\theta) + \log g_j(\theta_j) + \log g_{-j}(\theta_{-j})\right) d\theta_j d\theta_{-j} \\
&= -\int g_j(\theta_j)\left(\int g_{-j}(\theta_{-j})\log p(y,\theta)d\theta_{-j}\right)d\theta_j \\
&\quad + \int g_j(\theta_j)\log g_j(\theta_j)d\theta_j \\
&\quad + \int g_{-j}(\theta_{-j})\log g_{-j}(\theta_{-j})d\theta_{-j}
\end{aligned}$$

CAVI iteratively updates g_j for each j (similar to Gibbs sampler). For each g_j, the updates analogues EM.
E step: let $[g_j(\theta_j)]^t$ be the pdf of θ_j at the t-th iteration. For each j, only the first term depends on θ_j. Thus, we need to find the expected value

$$\mathrm{E}_{g^t_{-j}} \log p(\theta, y) = \int [g_{-j}(\theta_{-j})]^t \log p(y,\theta)d\theta_{-j}$$

M-step: the optimal update $[g_j(\theta_j)]^{t+1}$ is

$$\log[g_j(\theta_j)]^{t+1} = \mathrm{E}_{g^t_{-j}} \log p(\theta, y) + \text{const.},$$

or equivalently

$$[g_j(\theta_j)]^{t+1} \propto \exp[\mathrm{E}_{g^t_{-j}} \log p(\theta, y)]$$

In terms of algorithm, CAVI is similar to EM (or ECM). In principle, they are completely different.

Example: hierarchical normal model. Here, we consider a simpler setting, where $Y_j = \bar{Y}_{ij}$, $Y_j \sim N(\theta_j, \sigma_j^2)$, and $\theta_j \sim N(\mu, \tau^2)$, where σ_j^2 is assumed to be known in a priori. The log-posterior is:

$$\begin{aligned} \log p(\theta \mid y) &= -m \log \tau - \frac{1}{2\tau^2} \sum_{j=1}^{m} (\theta_j - \mu)^2 \\ &\quad - \frac{1}{2} \sum_{j=1}^{m} \frac{(y_j - \theta_j)^2}{\sigma_j^2} + \text{const.} \end{aligned}$$

Here, $\theta = (\theta_1, \cdots, \theta_m, \mu, \tau^2)$, and we want to approximate the posterior by the mean-field family.

For each θ_j:

$$\begin{aligned} \mathrm{E}_{g_{-\theta_j}} \log p(\theta, y) &= \mathrm{E}_{g_{-j}} \log p(\theta \mid y) + \text{const.} \\ &= -\frac{1}{2}\frac{(y_j - \theta_j)^2}{\sigma_j^2} - \frac{1}{2}\mathrm{E}(\tau^{-2})\mathrm{E}[(\theta_j - \mu)^2] + \text{const.}, \end{aligned}$$

implies

$$g(\theta_j) = N\left(\frac{\sigma_j^{-2} y_j + \mathrm{E}(\tau^{-2})\mathrm{E}(\mu)}{\sigma_j^{-2} + \mathrm{E}(\tau^{-2})}, \frac{1}{\sigma_j^{-2} + \mathrm{E}(\tau^{-2})}\right).$$

Similarly,

$$\mathrm{E}_{g_{-\mu}} \log p(\theta, y) = -\frac{1}{2}\mathrm{E}(\tau^{-2}) \sum_{j=1}^{m} \mathrm{E}[(\theta_j - \mu)^2] + \text{const.}$$

implies

$$g(\mu) = N\left(\frac{1}{m}\sum_{j=1}^{m} \mathrm{E}(\theta_j), \frac{1}{m}\frac{1}{\mathrm{E}(\tau^{-2})}\right),$$

and

$$g(\tau^2) = \text{Inv-}\chi^2\left(m-1, \frac{1}{m-1}\sum_{j=1}^{m} \mathrm{E}[(\theta_j - \mu)^2]\right)$$

Notice that the expectations can also be calculated explicitly.

To get variational Bayes started, we need to initialize the parameters in their distributions. We can draw the unbounded parameters from normal distributions and bounded parameters from Gamma or uniform distributions.

Other approximations: Expectation propagation (EP), Integrated nested Laplace approximation (INLA), Approximate Bayesian computation (ABC)...

11 Nonparametric models

11.1 Basis function mdoels

In general, a nonparametric regression model can be expressed as

$$Y = \mathrm{E}(Y|X = x) + \epsilon = \mu(x) + \epsilon$$

The name "nonparametric" indicates that the number of parameters required to specify $\mu(\cdot)$ is infinite. A common assumption is that $\mu \in L^2$, i.e., μ is square integrable:

$$< \mu, \mu >= \int |\mu(x)|^2 \, dx \leq \infty$$

Let $\{b_h(x)\}_{h=1}^{\infty}$ be the set of basis functions of the L^2 space, we can project μ

onto the basis:

$$\mu(x) = \sum_{h=1}^{\infty} \beta_h b_h(x).$$

We consider the following estimator of μ:

$$\hat{\mu}(x) = \sum_{h \in \mathcal{H}} \hat{\beta}_h b_h(x),$$

where $H = |\mathcal{H}| < \infty$ is a small integer.

No unbiased estimator for nonparametric models.

Basis functions: polynomials, Fourier series, *Gaussian radial basis, B-splines*
Gaussian radial basis:

$$b_h(x) = \exp\left(-\frac{|x - x_h|^2}{l^2}\right)$$

where x_h are centers of the basis functions and l is the bandwidth. This basis is motivated by kernel regression.

Cubic spline:

$$b_h(x) = \begin{cases} u^3/6 & \text{for } x \in (x_h, x_{h+1}), u = (x - x_h)/\delta \\ (1 + 3u + 3u^2 - 3u^3)/6 & \text{for } x \in (x_{h+1}, x_{h+2}), u = (x - x_{h+1})/\delta \\ (4 - 6u^2 + 3u^3)/6 & \text{for } x \in (x_{h+2}, x_{h+3}), u = (x - x_{h+2})/\delta \\ (1 - 3u + 3u^2 - u^3)/6 & \text{for } x \in (x_{h+3}, x_{h+4}), u = (x - x_{h+3})/\delta \\ 0 & \text{otherwise,} \end{cases}$$

where $\delta = |x_{h+1} - x_h|$ and $\{x_h\}$ is the set of knots.

In practice, we usually center the spline model to linear model

$$\mu(x) = \beta_1 + \beta_2 x + \sum_{h=3}^{H+2} \beta_h b_h(x)$$

After projecting the unknown function to H known basis functions, we can fit the model similar to fit a multivariate linear regression model.

The smoothness of $\hat{\mu}$ is controlled by the number of basis H. One can pre-specify a grid of many knots and then use either a variable selection prior (spike-and-slab prior) to effectively discard the knots that are not needed, or shrinkage prior (Bayesian ridge) to penalize the "roughness" of $\hat{\mu}$, e.g., smoothing splines and P-splines.

11.2 Gaussian process

Gaussian process is a flexible class of models for which any finite-dimensional marginal distribution is Gaussian, and which can be viewed as a potentially infinite-dimensional generalization of Gaussian distribution. Gaussian process is natural as a prior distribution for an unknown regression function $\mu(x)$.

The Gaussian process prior, denoted as $\mu \sim GP(m, k)$, parametrized in terms of a mean function m and a covariance function K. $m(x)$ is the mean function that represents an initial guess at the regression function. $k(x, x')$ is the covariance function that specifies the covariance between the process at any given points, which controls the smoothness of realizations from the Gaussian

process and the degree of shrinkage towards the mean.

For any $x_1, \cdots, x_n$, we have

$$\mu(x_1), \cdots, \mu(x_n) \sim N((m(x_1, \cdots, m(x_n)), K(x_1, \cdots, x_n)),$$

where $K(x_1, \cdots, x_n)$ is a $n \times n$ covariance matrix with element (i, j) corresponding to $k(x_i, x_j)$.

A convenient choice for the mean function is the zero function or linear model $m(x) = \beta_1 + \beta_2 x$, possibly with a hyperprior chosen for β. A common choice for k is the squared exponential (or exponentiated quadratic, or Gaussian) covariance function

$$k(x, x') = \tau^2 \exp\left(-\frac{|x - x'|^2}{2l^2}\right)$$

Here, τ controls the magnitude and l the smoothness of the function.

Gaussian process priors are appealing in being able to fit a wide range of smooth surfaces while being computationally tractable even for moderate to large numbers of predictors.

GP is closely connected to the basis expansion. Suppose

$$\mu(x) = \sum_{h=1}^{H} \beta_h b_h(x),\ \beta \sim N(\beta_0, \Sigma),$$

in terms of GP, we have

$$m(x) = b(x)\beta_0, k(x, x') = b(x)^\top \Sigma b(x),$$

where $b(x) = (b_1(x), \cdots, b_H(x))^\top$.

Different covariance functions can be used to add structural prior assumptions like smoothness, nonstationarity, periodicity, and multiscale or hierarchical structures. Sums and products of Gaussian processes are also Gaussian processes which allows easy combination of different covariance functions. Linear models can also be presented as Gaussian processes with a dot product covariance function.

One advantage of Gaussian process models over basis expansion methods is

that they can often be extended more naturally to multivariate regression settings. The mean function can be generalized to a multivariate linear model $m(x) = x^\top \beta$. Since it is typically not ideal to use a single parameter l to control the smoothness of μ in all directions, anisotropic GPs may be preferred and one can use, for example, a modified squared exponential covariance function

$$k(x, x') = \text{Cov}(\mu(x), \mu(x')) = \tau^2 \exp\left(-\sum_{j=1}^{p} \frac{|x_j - x'_j|^2}{2l_j^2}\right),$$

where $x = (x_1, \cdots, x_p)^\top$. One can do nonparametric variable selection by choosing hyperpriors for these l_j's, which controls smoothness in the direction of the j's predictor, to allow data adaptivity to the true anisotropic smoothness levels.

GP priors are appealing in being conditionally conjugate given (τ^2, l, σ^2), so that the conditional posterior for μ is a GP with updated mean function covariance process. In practice, we cannot estimate μ at infinitely many locations and hence the focus is on the observed data points $x = (x_1, \cdots, x_n)$ and any additional locations $\tilde{x} = (\tilde{x}_1, \cdots, \tilde{x}_{n'})$.

Given the prior $\mu \sim GP(0, k)$, we have

$$\begin{aligned}
\mu(\tilde{x})|x, y, \tau^2, l, \sigma^2 &\sim N(\text{E}(\tilde{\mu}), \text{Cov}(\tilde{\mu})) \\
\text{E}(\tilde{\mu}) &= K(\tilde{x}, x)(K(x) + \sigma^2 I)^{-1} y \\
\text{Cov}(\tilde{\mu}) &= K(\tilde{x}) - K(\tilde{x}, x)(K(x) + \sigma^2 I)^{-1} K(x, \tilde{x})
\end{aligned}$$

The major challenge is the computational burden for the inversion of the $n \times n$ matrix ($L \times O(n^3)$, where L is the number of MCMC draws). It becomes challenging to fit GP models with n is greater than a few thousand and the number of predictors is greater than 3.

Approximation methods are proposed, with additional assumptions made on $k(x, x')$, to speed up the matrix inversion burden, e.g., Markov random field ($O(n)$ for univariate x and $O(n^{3/2})$ for spatial problems) and basis function approximation